MC90HDNC1L/R Integrated Cabinet Air Conditioner

MC90HDNC1L/R 一体式机柜空调

User Manual

用户手册

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About This Document

Purpose

This is a general version user manual for MC90HDNC1L/R cabinet air conditioners. It mainly introduces the product overview, installation, controller operation, routine maintenance and troubleshooting of the cabinet air conditioner which is convenient for readers to master the use and maintenance of this series air conditioner.

Intended Audience

This document is mainly applicable to the following readers:

- Product installation engineer
- Technical support engineer
- After-sales maintenance engineer

Symbol Conventions

The following signs may appear in the text, with the following meanings.

Symbol	Description
	Warns of a critically hazardous situation, which may result in death or serious personal injury if the requirements are not followed.
	Warns of a potentially dangerous situation, which may result in moderate or minor personal injury if the requirements are not followed.
	Warn of scenarios that affect equipment safety or environmental safety. Failure to comply with the requirements may cause damage to equipment or other adverse results.
	Prompts for some general details, or other useful information to simplify the work.

Change History

Changes between document issues are cumulative. The latest document issue contains all the changes made in earlier issues.

Version	Update	Modified Date
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1 Safety Precautions

Please read this chapter carefully before carrying out assembly, installation, electrical engineering and maintenance operations, and strictly abide by relevant requirements and operate carefully when performing relevant operations.

Declaration

Envicool does not take responsibilities for the following situations:

- Operate the device in abnormal conditions or environments that do not meet the application scenarios described in this manual.
- Failure to comply with the operating requirements of this manual, equipment failure due to improper use, or incorrect installation and maintenance methods.
- Failures not caused by this equipment (such as failures caused by the customer's equipment, software, etc.).
- Device damage caused by yourself replacement or disassembly of product parts, or device damage caused by unauthorized maintenance service personnel.
- Device damage caused by force majeure such as fire, earthquake, flood, etc.

General Safety

- Any personal injury caused by misoperation should be dealt with immediately, and first aid or medical care must be obtained immediately.
- Please strictly abide by the local safety regulations and laws and regulations to avoid accidents. The safety precautions in the manual do not represent all the safety precautions that should be followed, but only serve as a supplement to local safety regulations.
- If there are warning labels attached to the device, please read them carefully and strictly observe them before operation.
- Only original parts and parts authorized by the manufacturer to ensure the stability and safety of air conditioner device are allowed to be used. If other parts are used, the warranty right will be invalid.
- It is not allowed to make any changes to air conditioner devices without the agreement and consent of the manufacturer.
- Children and personnel with limited cognitive and coordination ability are not allowed to operate, maintain, clean the device or use it as a toy.
- Please entrust professionals for installation, electrical engineering and maintenance. Self-installation may lead to accidents such as water leakage, electric shock or fire.



Professionals refer to those who have professional training, experience and skills, and are familiar with regulations, rules and measures related to accident prevention and operation requirements, and have the ability to discover and avoid all potential hazards. The operator or the person in charge authorizes this person to ensure the safety of the equipment and carry out all necessary operations.

Electrical Safety

- It is forbidden to install or remove the power cord when powered on. Please turn off the power switch before installing or removing the power cord.
- When installing the device, always make the ground connection first and disconnect it at the end. Do not power on the device in the absence of a properly installed ground conductor.
- Please perform installation and maintenance operations after the power is off, and do not turn on the power during the air conditioner device is running. For some operations that need to be performed during running (such as measuring current, pressure, temperature, etc.), you must ensure that the device wiring has been completed correctly before the power can be turned on, and the power should be cut off immediately after the operation is completed.



The capacitors contained in the air conditioner will continue to discharge after the power is cut off. To ensure personal safety, please wait at least 1 minute after the unit is powered off before performing maintenance operations.

- Protective measures must be taken during electrical maintenance, such as wearing insulating gloves or insulating shoes.

Operational Safety

- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.
- Children should be supervised to ensure that they do not play with the appliance.
- The appliance shall be installed in accordance with national wiring regulations.
- When the unit is running, please do not touch the internal components of the unit with your hands at will to avoid electric shock or injury by the fan blades of the fan.
- If severe vibration or abnormal sound occurs during running or debugging of the unit, please stop all operations and immediately cut off the circuit switch for inspection.
- All circuit switches should be disconnected during inspection and maintenance, and guards should be set up to prevent others from closing the circuit switches by mistake.
- Only when all the circuit switches are turned off and the internal control board no longer flashes the signal light, can you operate the device circuit and electronic devices, and you must wear anti-static gloves.
- Do not allow liquids such as water to enter the terminal area of the device during installation and maintenance.
- It is forbidden to carry heavy device by hand alone. Please wear protective gloves when carrying light device by hand.
- Please wear protective gloves when there are sharp objects (such as fins, etc.) in the operating parts.
- Be careful when cutting the cable tie to avoid cutting the cable.

Application Safety

- Do not use this product in a flammable and explosive environment.
- This series of cabinet air conditioners are outdoor air conditioners. If the air conditioners need to be used indoors, please confirm with the product manufacturer and implement necessary waterproof measures.
- Please be cautious when professional maintenance is involved. For details, please consult the customer service center of the product manufacturer.

Recycling



It is strictly forbidden to dispose of this product as unsorted municipal waste, please sort it and recycle it. The disposal of scrap must comply with relevant requirements and adopt special methods for independent treatment to avoid potential pollution to the environment.

2 Product Description

2.1 Feature

2.1.1 Product Introduction

Model Description

The naming rules of MC series cabinet air conditioners are shown in **Table 2-1**.

Table 2-1 Naming rule

MC	90	H	D	N	C	1	L/R
1	2	3	4	5	6	7	8

1 Product series	MC: energy storage integrated air conditioner series
2 Rated cooling capacity	90: 9kW
3 Basic function	H: with heating
4 Installation method	D: embedded in the door
5 Application environment	N: normal temperature
6 Noise level	C: common noise
7 Input Power supply	1: Single-phase AC input
8 Version code	L or R

Feature Description

MC90HDNC1L/R integrated cabinet air conditioner is a refrigeration product developed for application environments such as cabinets or containers, and is applied to the following scenarios:

- The devices in the cabinet generate a large amount of heat.
- The devices in the cabinet are sensitive to ambient temperature.
- The devices in the cabinet must be completely isolated form the outside.

MC90HDNC1L/R integrated cabinet air conditioners mainly have the following features:

Table 2-2 MC series units features

Feature	Description
Advanced design	• It adopts the integrated design, which is simple and convenient to install. It can work after being powered on, without complex commissioning. • It adopts outdoor mental structure and anti-corrosion measures.

Feature	Description
Complete functions	• It supports smart refrigeration, heating and dehumidication functions. • It supports RS485 modbus communication function. • It supports power-off memory and self-start function.
Stable performance	It can keep running 24 hours a day, uninterrupted throughout the year.
Energy efficient	It adopts AC power supply, equipped with inverter compressor and EC fan.

2.1.2 Function Description

Smart Refrigeration / Heating / Dehumidication

The unit is equipped with an high-efficiency controllers, and a display is provided for viewing the unit status and setting parameters, which is easy to operate.

The controller monitors the return air temperature and humidity in the cabinet in real time, and makes intelligent adjustment according to the target temperature and humidity setting, and realizes the smart cooling, heating and dehumidication functions by controlling the operation of parts such as compressor, fans and electric heaters.

RS485 Modbus Communication

The unit communicate with the host computer through the RS485 interface using modbus protocol. By sending request messages, the host computer can perform operations such as querying the operating status of the unit and setting control parameters to realize remote monitoring of the unit.

Power-off Memory

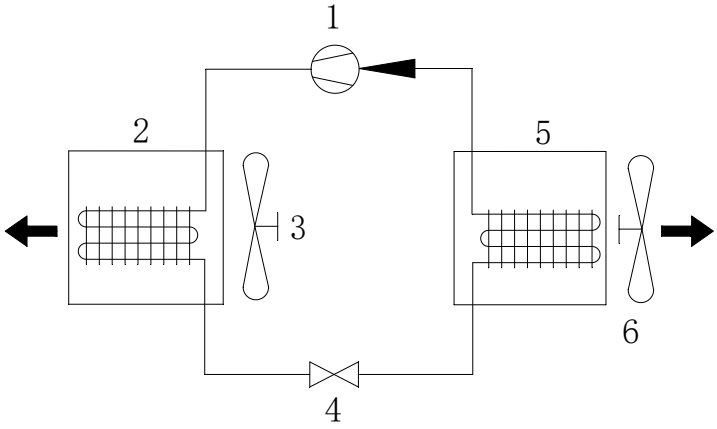
After the unit is powered off, it can memorize the settings before power off, and it can automatically and intelligently run according to the memorized settings after the power is turned on again.

2.2 Implement

2.2.1 System Principle

MC series cabinet air conditioners are mainly composed of compressor, condenser, external circulation fan, evaporator, internal circulation fan and control box. The realization principle of system refrigeration is shown in **Figure 2-1**.

Figure 2-1 System implement principle



1: Compressor 2: Condenser

3: External circulation fan 4: Throttle element

5: Evaporator 6: Internal circulation fan

- **Compressor**: It is responsible for compressing the refrigerant and providing power for the cooling system.
- **Condenser**: It adopts finned tube design, which has high heat exchange efficiency. The refrigerant condenses and releases heat in the condenser.
- **Throttling element**: It adopts an electronic expansion valve, which is used to control the refrigerant flow by adjusting the opening..
- **Evaporator**: It adopts finned tube design, which has high heat exchange efficiency. The refrigerant evaporates and absorbs heat in the evaporator.
- **Indoor fan**: It is responsible for blowing indoor air across the surface of the evaporator coil for heat exchange.
- **Outdoor fan**: It is responsible for discharging the heat released when the refrigerant in the condenser coil is condensed to the outside.
- **Control box**: It performs logical control on the parts of the air conditioner to achieve temperature control requirements.

2.2.2 Working Process

The working process of MC series integrated cabinet air conditioner is as follows:

1. When the internal temperature of the cabinet or container reaches the cooling opening point, the compressor starts, and the compressor compresses the refrigerant.

   After being compressed by the compressor, the refrigerant is transformed into high-pressure and high-temperature gas and enters the condenser.

2. The condenser condenses the gaseous refrigerant into a liquid state.

   The outdoor fan takes away the heat released when the condenser condenses the refrigerant, and the refrigerant enters the throttling element.

3. The throttling element reduces the pressure and temperature of the refrigerant.
4. Low temperature and low pressure refrigerant wet vapor enters the evaporator to absorb heat.

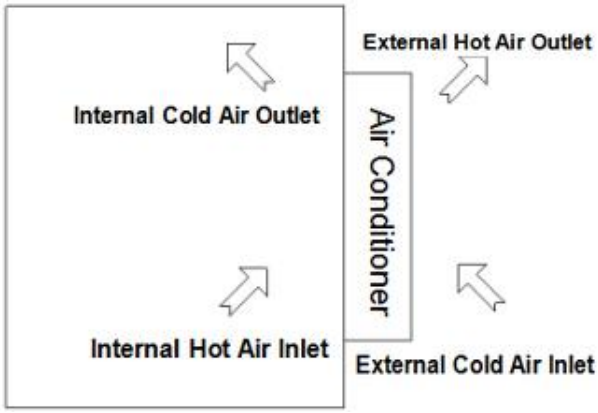
The indoor fan takes away the heat of the internal space such as cabinets or containers to achieve the cooling effect.

5. After the refrigerant wet vapor passes through the evaporator to absorb heat, it becomes a low-pressure and low-temperature gas again that enters the compressor for compression and circulates for refrigeration process.

2.2.3 Air Circulation Design

The air circulation system of the MC series integrated air conditioner is shown in **Figure 2-2**.

Figure 2-2 Air circulation diagram



- In the internal circulation air path, the air conditioner sucks in the hot air in the cabinet from the lower part, and discharges the cold air from the upper part after heat exchange.
- In the external circulation air path, the air conditioner sucks in outdoor cold air from the lower part, and discharges the hot air from the upper part after heat exchange.

2.2.4 Control Logic

The control logic of MC series integrated air conditioner is shown in **Table 2-3**.

Table 2-3 Temperature control logic

Item	Control Logic	Description
Cooling	• Cooling opening condition: cooling demand ≥ 50% • Cooling stop condition: cooling demand ≤ 0%	Cooling demand = (indoor return air temperature - cooling set point [23°C]) / cooling sensitivity [5°C] * 100%
Heating	• Heating opening condition: heating demand ≥ 100% • Heating stop condition: heating demand < 0%	Heating demand = (heating set point [15°C] - indoor return air temperature) / heating sensitivity [5°C] * 100%
Dehumidification	• Dehumidification opening condition: dehumidification demand ≥ 100% • Dehumidification stop condition: dehumidification demand < 0%	Dehumidification demand (%) = (indoor return air humidity - dehumidification setting point [default 60%]) / dehumidification sensitivity [default 15%]* 100%

Table 2-4 Unit parts work logic

Unit Parts	Working Logic
Compressor	The default cooling set point of the unit is 23°C, and the cooling sensitivity is 5°C. Under default conditions, the cooling start temperature is 25.5°C (23°C+5°C*50%=25.5°C), and the cooling stop temperature is 23°C (23°C+5°C*0%=23°C). The default dehumidification set point for the unit is 60%, with a return difference of 15%. Under default conditions, the dehumidification start humidity is 75% (60%+15% * 100%=75%), and the dehumidification stop with a humidity of 60% (60%+15% * 0%=60%); • When the indoor temperature is higher than 25.5°C, the compressor starts to run. • When the indoor temperature is lower than 23°C, the compressor stops running. • When the indoor return air humidity is higher than 75% of the dehumidification opening point, the compressor starts running. • The compressor needs to wait three minutes before restarting after each shutdown to avoid compressor fault due to frequent start and shutdown.
Indoor fan	The working mode of the indoor fan after the air conditioner is turned on is as follows: • Air supply mode: When the indoor temperature is lower than 23°C, the air conditioner works in air supply mode, and the indoor fan keeps running at low speed. • Cooling mode: The indoor fan runs at high speed. • Dehumidification mode: The indoor fan runs at low speed.
Outdoor fan	• The outdoor fan adjusts its speed according to the condensing pressure. • When the condensing pressure is low, the external circulation fan reduces the speed through variable speed control to achieve the purpose of reducing noise.
Electric heater	The default heating set point of the unit is 15°C, and the heating sensitivity is 5°C. Under default conditions, the heating start temperature is 10°C (15°C-5°C*100%=10°C), and the heating stop temperature is 15°C (15°C-5°C*0%=15°C). • When the indoor return air temperature is lower than 10°C, the electric heaters start to operate. • When the indoor return air temperature is higher than 15°C, the electric heaters stop running.

2.3 Specification

The specification parameters of the MC integrated air conditioner are shown in **Table 2-5**.

Table 2-5 Parameter specification

Parameter	Specifications
Operating voltage range	187VAC ~ 253VAC 50/60Hz
Operating environment temperature range	-30°C~55°C
Operating environment relative humidity range	5% ~ 95%
Storage environment temperature range	-40°C~75°C
Storage environment relative humidity range	5% ~ 95%
Transport	Adapted to transportation by train, car, airplane, sea, etc.
Unit size (H×W×D)	1770mm×620mm×300mm

NOTE

The technical parameters marked on the equipment nameplate will not be repeated here.

3 Installation Guide

3.1 Installation Instructions

In order to achieve the design performance of units and maximize their service life, please reserve space for installation and maintenance with reference to the suggestions provided by manufacturers, and strictly install equipment drawings for installation.

3.1.1 Tools Preparation

Before installation and maintenance, please prepare not limited to the commonly used tools listed in **Table 3-1**. On-site technicians can increase or decrease them according to the actual situation.

Table 3-1 Tools preparation

Tool	Illustration	Function
Flat-head screwdriver (2mm×75mm)		A flat-blade screwdriver installs or removes the connecting wires of the controller, etc.
Phillips screwdriver (M3/M4/M6/M8)		Phillips screwdriver is used to install or remove screws for other parts.
Diagonal pliers		Process the cables and remove the cable ties on site.
Clamp ammeter		• Check the working voltage and current on site. • Check the internal power distribution voltage and current.
Tape measure		Measure the size on site.
Crimping pliers		On-site processing of cables and crimping of various terminals.
Special flushing tool		Use a water gun to clean the condenser.
Brush		Clean up dust.

3.1.2 Transportation and Handling

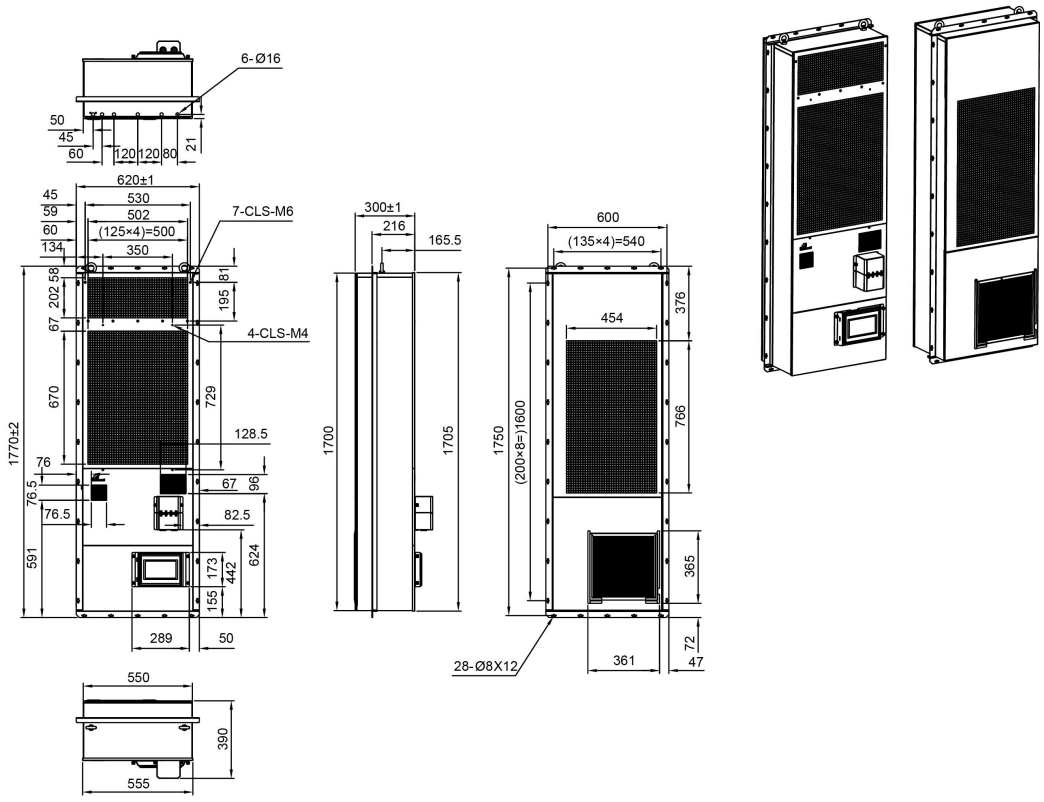
- During transportation and handling, please ensure that the outer packaging is complete, avoid moisture or damage to the outer packaging, and ensure that the unit is protected from collision, rain and chemical corrosion.

- Before unpacking, move the air conditioner to the nearest place to the installation site as far as possible.
- During transportation and placement, the air conditioner must be kept upright.

3.1.3 Physical Dimension

The dimension of the MC90HDNC1L/R Integrated Cabinet Air Conditioner is shown in **Figure 3-1**.

Figure 3-1 Dimension of the MC90HDNC1L/R

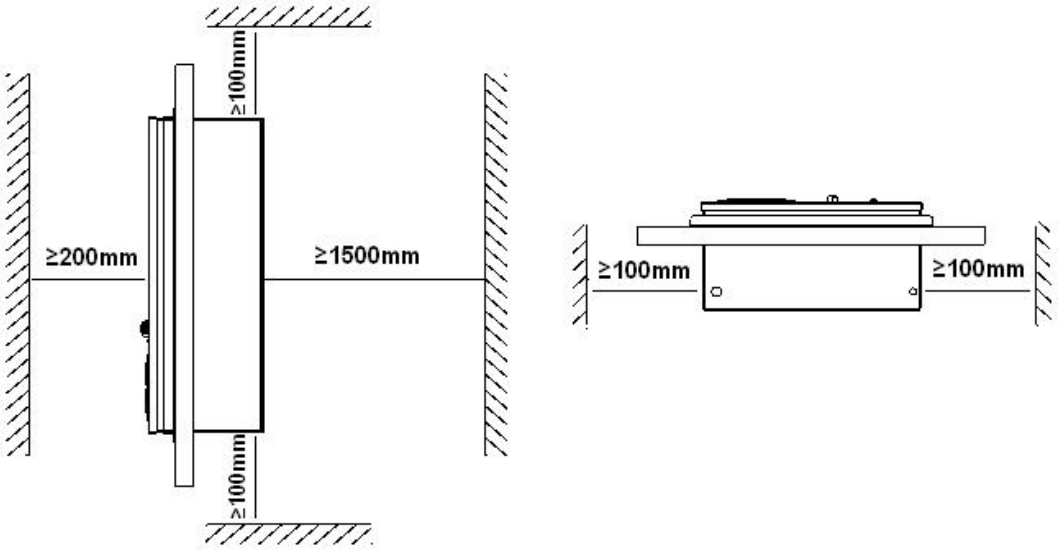


3.1.4 Installation layout

In order to ensure the normal operation of the unit, no objects are allowed to be placed at the internal and external circulation air inlets and outlets to obstruct the air flow.

Please refer to **Figure 3-2** to install the air conditioner on a fixed device (such as a cabinet door).

Figure 3-2 Installation layout diagram



The specific requirements for the installation and maintenance space of the unit are as follows.

Table 3-2 Installation and maintenance space requirements

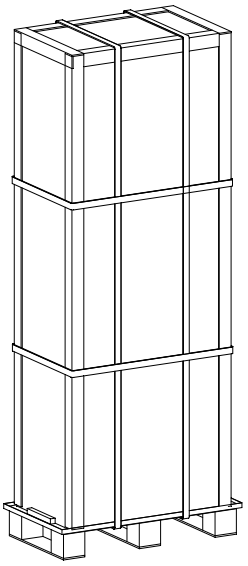
Item	Requirements
Indoor maintenance space	Avoid installing or placing any device at least 200mm in front of the air inlet and outlet of the internal circulation fan, otherwise the air path may be obstructed.
Outdoor maintenance space	Avoid installing or placing any device at least 1500mm in front of the air inlet and outlet of the external circulation fan, otherwise, the air path may be obstructed.

3.2 Unpacking and Inspection

Before unpacking and disassembling, please use lifting equipment to move the machine as close as possible to the installation location to facilitate the movement of the unit. At the same time, ensure that the unit is placed upright, and it is not allowed to lay flat or lie on its side.

The product is packaged in cartons packing. When unpacking, please remove the packing straps on the outside of the package first, and then open the cartons packing, as shown in **Figure 3-3**.

Figure 3-3 Unpacking diagram



If the product is not installed immediately, or the product needs to be transferred to other regions, please repackage the unit after the unpacking inspection is completed.

Please do the following checks after unpacking:

- Please carefully check whether the unit is damaged due to transportation, and pay particular attention to whether there are loose parts, pits, scratches, and oil leakage. If any damage is found, please report it to the seller with the barcode number on the outside of the package within one week.
- Please count the number of units and installation accessories according to the list of accessories.
- Before disposing of the packaging materials, please make sure that there are no loose parts in the packaging materials. In order to protect the environment, it is recommended to recycle the packaging boxes.

3.3 **Mechanical Installation**

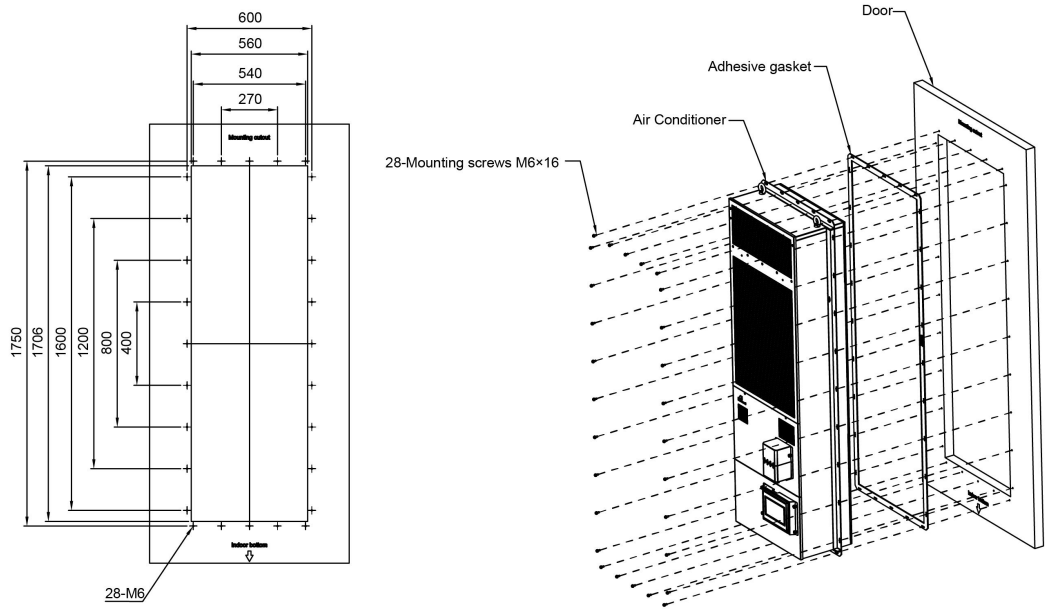
Prerequisites

- The installation location of the unit has been reasonably planned by referring to **3.1.3**Physical Dimension and **3.1.4**Installation layout.
- The installation tools and required accessories are ready.

Context

Please refer to **Figure 3-4** for the installation of MC90HDNC1L/R unit.

Figure 3-4 Installation diagram



Procedure

Step 1 Move the unit to the planned installation location.

Step 2 Use a Phillips screwdriver and screws to fix the unit on the cabinet door.

Step 3 Connect the condensate pipe to the reserved drain pipe.

3.4 Electrical Connection

Prerequisites

Mechanical installation has been completed.

Precautions

WARNING

- Electrical connections must be operated by trained professionals.
- When connecting the power cord, live operation is prohibited.
- Ensure that the input power supply meets the requirements of the unit. Please refer to the product nameplate parameters for details.
- In order to avoid any destructive influence, the unit must be grounded reliably before turning on the power supply.
- All electrical connections must meet the requirements of national and local electrical codes.

Procedure

Step 1 Connect the power cords.

The power interfaces are shown as **Figure 3-5**. Please refer to **Table 3-3** for interfaces definition.

Figure 3-5 Power cord interfaces

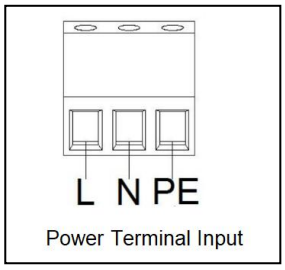


Table 3-3 Power interface pins definition

Pins	Definition	Description
L	Single-phase power supply L (L1)	The recommended power cord specification is 3×10mm² （3×8AWG）, and only copper cables can be used.
N	Single-phase power supply N (L2)	
PE	Single-phase power supply PE	

Step 2 Connect the communication lines.

The communication interfaces are shown as **Figure 3-6**. Please refer to **Table 3-4** for interfaces definition.

Figure 3-6 Communication interfaces

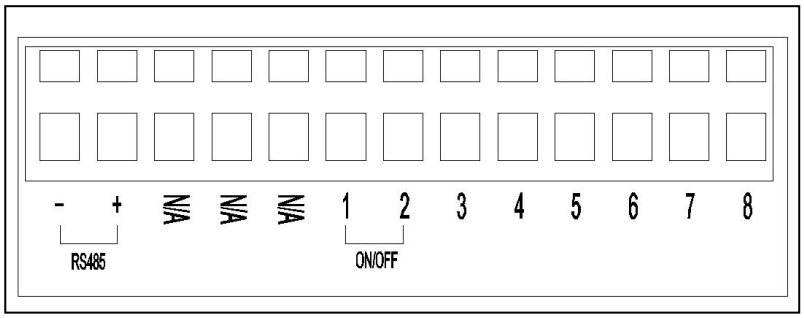


Table 3-4 Communication interface pin definition

Pin	Pin	Definition	Description
Communication interface pin	RS485-	RS485-	Host computer monitoring interfaces, recommended to use 2×0.5mm² twisted-pair shielded network cables.
	RS485+	RS485+	
	N/A	-	-
	N/A	-	
	N/A	-	

Pin	Pin	Definition	Description
	1	Remote control switch pin -	Recommended to use 2×0.5mm^2 twisted-pair shielded network cables.
	2	Remote control switch pin +	
	3	-	-
	4	-	-
	5	-	-
	6	-	-
	7	-	-
	8	-	-

Step 3 Connect the 5 PIN display terminal block to the machine.

Step 4 After the connection is completed, use a clamp ammeter to check whether there is a short circuit or open circuit in the power line and signal line.

3.5 Check After Installation

After the installation is complete, please refer to the information listed in **Table 3-5** to check again. Only when all checks are passed, the power on is allowed.

Table 3-5 Check after installation

Item	Standards
Unit installation	The units are placed horizontally and the fastening parts are locked.
	The condensate pipe has been connected to the drain pipe.
	Sundries (such as transportation materials, structural materials, and tools) inside or around the units have been removed.
Electrical connections	The power supply voltage is the same as the rated voltage on the device nameplate.
	All cables and circuit connectors have been tightened, and the tightening screws are not loose.
	There is no damage to the cable.
	There is no open circuit, short circuit and wrong connection in the electrical circuit of the system.

4 Controller

4.1 Introduction

The air conditioner of MC90HDNC1L/R unit support external handle operator to provide users with a human-computer interaction interface. Users can query, set up and monitor conveniently through the touch screen to ensure the normal operation of the chiller.

4.2 Control System Interface

Turn on the power switch to power on the chiller, and enter the control system homepage after the system is initialized, as shown in **Figure 4-1**.

Figure 4-1 Control system homepage interface



After the system is powered on, press any key and the backlit light will light up. If there is no keyboard operation after a period of time, the backlit light will dim.

The meanings of icons in the control interface of the chillers are shown in **Table 4-1**.

Table 4-1 Control interface icons

Icon	Description
	Switch button icon, click it to change the current working state of chiller.
	Language icon, click it to switch the language of the current interface to **Chinese** or **English**.
	Working status icon, which displays the current working status of chiller in the form of indicator lights. • Green indicates running status. • Red indicates the shutdown state.

Icon	Description
	Alarm icon, which indicates that the current chiller has generated alarms.
	Cooling icon, which indicates that the chiller is in cooling state, and the compressor is working.
	Heating icon, which indicates that the chiller is in heating state, and the heater is working.
	System homepage icon, click it to enter the operation homepage interface of the control system, you can view the current time, operation mode, return air temperature, humidity and interface language of the unit.
	System status icon, click it to enter the **System Status** interface, you can view the environment status, running status and group control status.
	System alarm icon, click it to enter the **Alarm** interface, you can view current alarms and historical alarms.
	System operation record icon, click it to enter the **Record** interface, you can view the chiller's operation record, operation log and operation curve.
	System setting icon, click it to enter the **Parameter Setting** interface,you can set the state of health mode, humidification and drainage.
	System maintenance icon, click it to enter the **Maintain** interface, you can manually start the coolant replenishment function, and set the replenishment related time parameters.

4.3 System Start-up and Shutdown



- After the chiller is powered on, if the compressor does not start, the heating belt will automatically start heating. When the chiller is powered on for the first time, it is necessary to ensure that the heating time of the compressor heating belt is not less than 4 hours before starting up the chiller.
- When the chiller has not been turned on for a long time (about 4 months), the chiller must be powered on to make the compressor heating belt work (the circulating water pump should not be started). The refrigerant liquid accumulated in the compressor will be heated and volatilized, otherwise direct start-up will have an adverse effect on the compressor. The heating time of the compressor heating belt must not be less than 10 hours.
- When starting the chiller at low temperature, the compressor will produce abnormal situations that are difficult to start and poor loading/unloading due to the excessive viscosity of the lubricating oil. Therefore, it is necessary to preheat the compressor through the compressor heating belt in advance (heating for not less than 4 hours). Do not cut off the power when the chiller is shut down to ensure that the compressor heating belt is continuously heated.

After clicking the switch button every time, the system switches the working state of the current chiller.

The dialog box that pops up when the chiller is start up is shown in **Figure 4-2**.

Figure 4-2 System start-up



After starting, the chiller will be updated from the shutdown state to the running state. On any interface, if there is no keyboard operation for a period of time, the system will automatically return to the home page interface.

The dialog box popped up when shutting down is shown in **Figure 4-3**.

Figure 4-3 System shutdown



After shutting down, the chiller will be updated from running status to shutdown status.

The system supports power-off memory function. If the system is powered on after power off, the system will automatically enter the state before power off (for example, if the system was on before power off, the system will automatically start up and enter the normal display interface after power on).

4.4 Status

Click **Status** on the homepage interface of the system to view the current environmental status, operating status and group control status (optional) of the chiller.

Figure 4-4 Environment Status interface

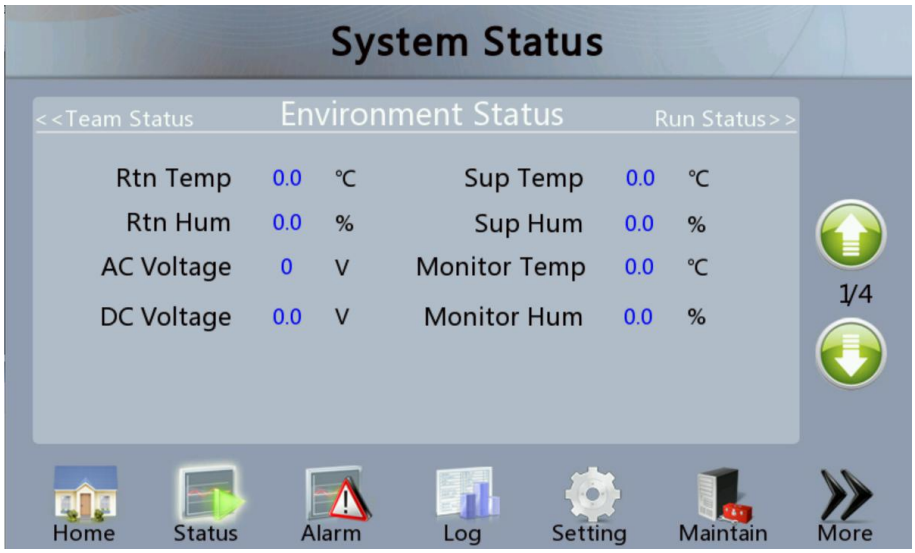


Figure 4-5 Run Status interface

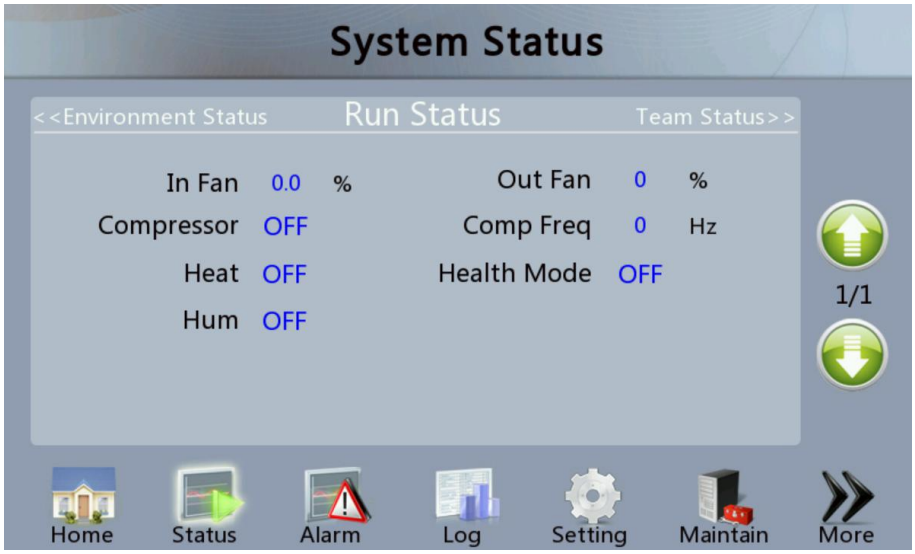
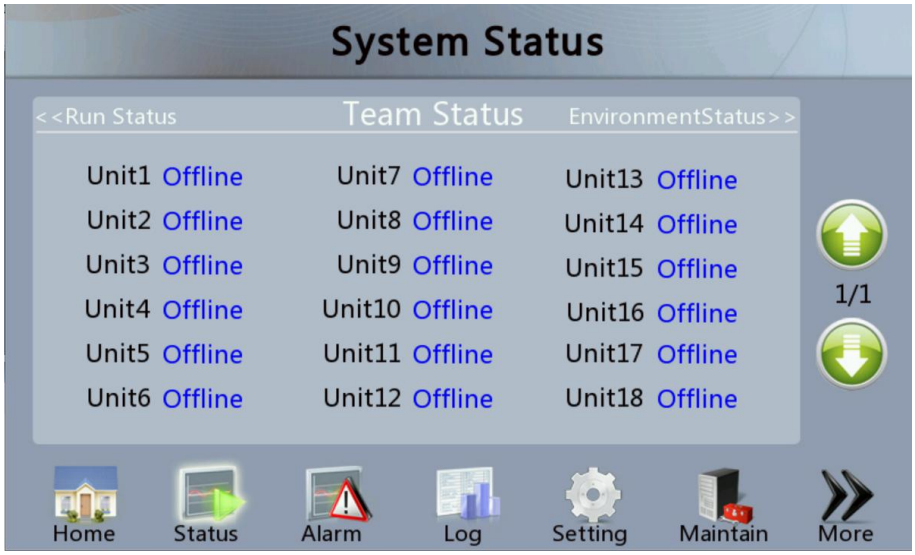


Figure 4-6 Team Status interface



4.5 Alarm

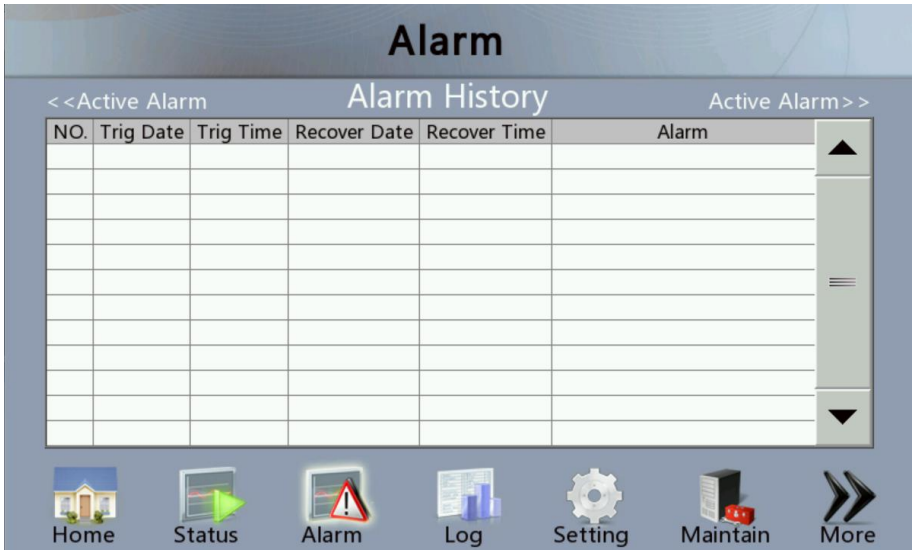
Click **Alarm** on the homepage interface of the system to view the current active alarm and historical alarm information of the chiller.

Figure 4-7 Active Alarm interface



You can reset the current alarm by clicking **Options** > **Reset Alarms**.

Figure 4-8 Alarm History interface



4.6 Record

Click **Record** on the homepage interface of the system to view the current operation record, operation log of the chiller and operation curves.

Figure 4-9 Run Time interface

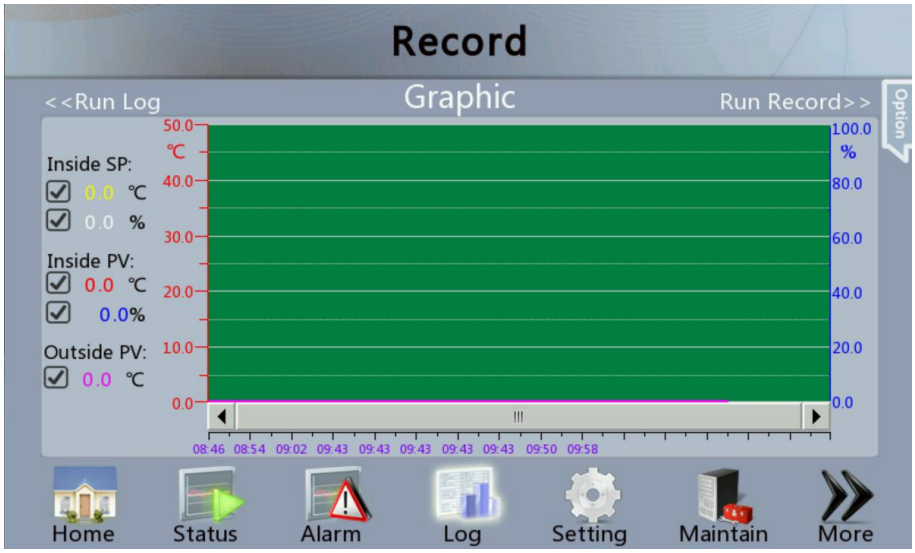


When a chiller leaves the factory, the initial values of the running time and the number of actions are **0**, and their values are automatically accumulated according to the actual situation.

Figure 4-10 Run Log interface



Figure 4-11 Graphic interface



The system displays the **Inside SP**(refrigeration set value, dehumidification set value) curve, **Inside PV**(actual return air temperature, actual humidity) curve, and **Outside PV** (outdoor actual temperature) curve. The user can perform the following operations in the "**Graphic** " interface

- Customize the curve to be displayed by unchecking the check box in front of the curve type.
- Click **Options** > **Clear Graphic** to clear the curve displayed on the current interface.

  After the curve is cleared, the control system automatically regenerates the curve according to the operating situation of the chiller.
- Click the temperature value and pressure value around the curve display area, you can modify the temperature value and humidity value range according to the actual plan.

4.7 Setting

Temp Setting

CAUTION

- In order to make the chiller run reliably and maximize energy efficiency, it is recommended that users do not change the temperature setting point at will.
- When the parameter setting value does not meet the actual operation requirements and needs to be modified, please confirm with the product after-sales service.

Click **Setting** on the homepage interface of the system and select the **Temp&Hum Setpoint** tab. The system displays the interface shown in **Figure 4-12**, and the parameter settings of the interface are shown in **Table 4-2**.

Figure 4-12 Temp&Hum Setpoint interface

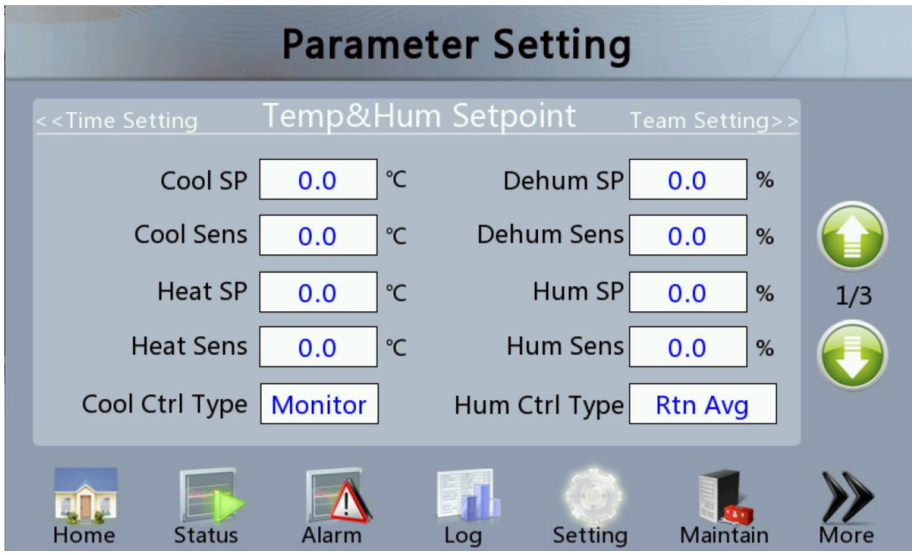


Table 4-2 Parameter description of the **Temp&Hum Setpoint** interface

Parameter	Default Value	Setting Range	Unit	Description
Cool SP	23	[7~50]	°C	Refer to **Table 2-3** for the opening and closing conditions of refrigeration, heating and dehumidification.
Cool Sens	5	[1~15]	°C	
Heat SP	15	[-40~20]	°C	
Heat Sens	5	[1~15]	°C	
Dehum SP	60	[10~90]	%	
Dehum Sens	15	[3~20]	%	

Parameter	Default Value	Setting Range	Unit	Description
Control Temp	Rtn Avg	• Rtn Avg • Sup Avg • Rtn Hot • Sup Hot • Monitor	-	Identifies the type of temperature controlled by the system. • **Rtn Avg**: The system controls according to the average return air temperature. • **Sup Avg**: The system controls according to the average value of the air outlet temperature. • **Rtn Hot**: The system controls according to the highest return air temperature. • **Sup Hot**: The system controls according to the highest value of the air outlet temperature. • **Monitor**: The system controls according to the cell temperature sent by the host computer during communication.
Control Hum	Rtn Avg	• Rtn Avg • Sup Avg • Tea Rtn Avg • Tea Sup Avg • Monitor	-	Identifies the type of temperature controlled by the system. • **Rtn Avg**: The system controls according to the average return air humidity. • **Sup Avg**: The system controls according to the average value of the air outlet humidity. • **Tea Rtn Avg**: The system controls according to the average return air humidity of multiple units under group control. • **Tea Sup Avg**: The system controls according to the average value of air outlet humidity of multiple units under the condition of group control. • **Monitor**: The system controls according to the cell humidity sent by the host computer during communication.

NOTE

- The default values and value ranges of the above parameters are for reference only, and the factory settings shall prevail.
- The parameters not listed in the table remain factory default values.

System Setting

Click **Setting** on the homepage interface of the system and select the **System Config** tab. The system displays the interface shown in **Figure 4-13**, and the parameter settings of the interface are shown in **Table 4-3**.

Figure 4-13 System Config interface

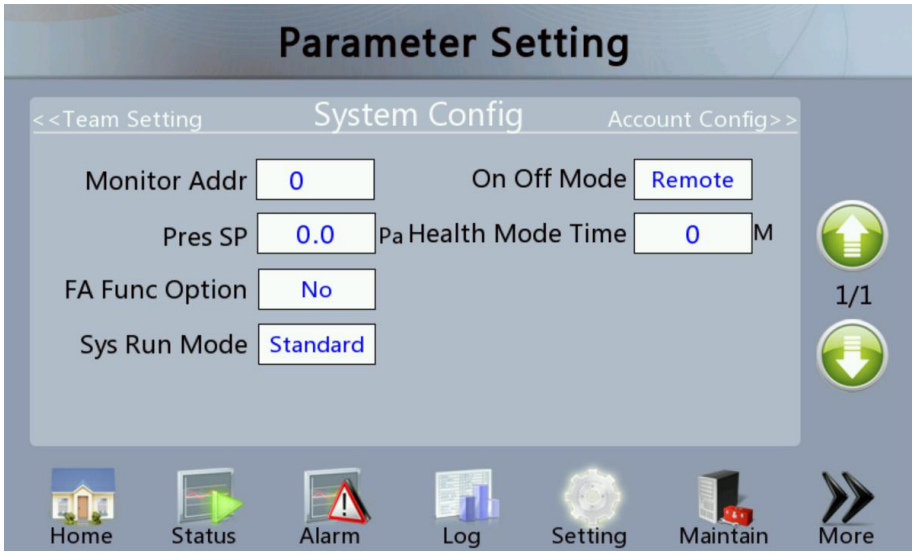


Table 4-3 Parameter description of the **System Config** interface

Parameter	Default Value	Setting Range	Description
Monitor Addr	1	[1~247]	The monitoring address of water chillers is unique when there are multiple units.
On Off Mode	Remote	• Local • Remote	The air-cooled chiller supports the following two switch modes: Button switch, monitoring switch (Modbus protocol switch) and remote switch (dry contact switch). • **Local**: The monitoring switch mode is invalid.When shutting down by press the key, the chiller enters the off state. When both switch modes are in the on state, the chiller enters the on state. • **Remote**: When shutting down in any of the three switch modes, the chiller enters the off state. When

Parameter	Default Value	Setting Range	Description
			all three switch modes are in the on state, the chiller enters the on state.

NOTE

- The default values and value ranges of the above parameters are for reference only, and the factory settings shall prevail.
- The parameters not listed in the table remain factory default values.

Team Setting

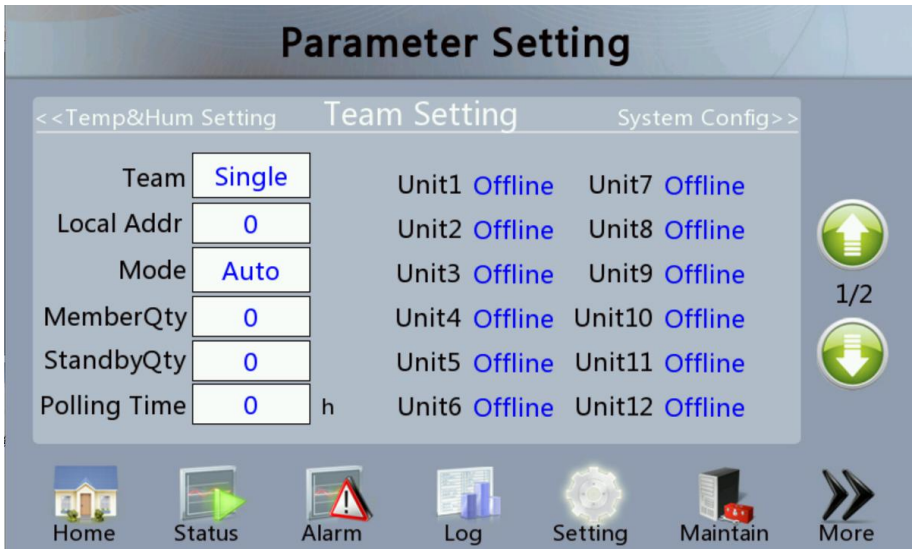
CAUTION

When supporting the group control function, the operator must make the relevant group control setting after the chiller is powered on for the first time and runs normally, otherwise the group control function will not be realized, and the chiller will return to the independent operation control mode.

Group control can improve the efficiency of chillers and avoid the competition between chillers by cascading multiple chillers.

When the group control is offline, the parameter value in the configuration interface is "offline", as shown in **Figure 4-14**.

Figure 4-14 Team Setting interface



When the chiller provides group control function, you can refer to **Table 4-4** for parameter setting.

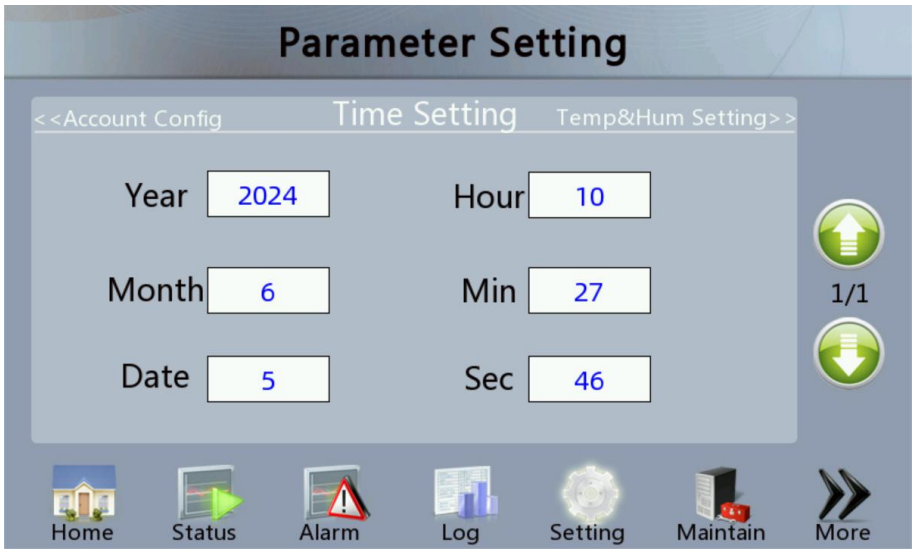
Table 4-4 Parameter description of the **Team Setting** interface

Parameter	Description
Team	Set to **Team** when supporting group control function.
Local Addr	The address of this machine and the address of the main control unit are configured as **1**. If the number of group control units is 2, that is, when two chillers back up each other, the local address is configured as **1**, and the other backup unit address is configured as **2**.
Mode	Set to **Auto**.
MenberQty	Number of groups. If two chillers backup each other, the configuration is **2**.
StandbyQty	Number of standby machines. If two chillers backup each other, the configuration is **1**.
Polling Time	Round patrol time for the system to exchange master and standby. Default value: 24 Unit: Hours
Team Ctrl	Temperature type used in group control. Set to **Team Avg**.

Time Setting

Click **Setting** on the homepage interface of the system and select the **Time Setting** tab. The system displays the interface shown in **Figure 4-15**. The current system time of the unit can be set in this interface.

Figure 4-15 Time Setting interface



Account Config

Click **Setting** on the homepage interface of the system and select the **Account Config** tab. The system displays the interface shown in **Figure 4-16**.

Figure 4-16 Account Config interface



The operator can modify the customer password through the account setting function. The default customer password is **0001**, it is recommended to modify it after logging in to the control system for the first time.

4.8 Maintain

Click **Maintain** on the homepage interface of the system, and the system displays the interface shown in **Figure 4-17**.

Figure 4-17 Maintain interface



The MC90HDNC1L/R Integrated Cabinet Air Conditioner does not support the current maintenance interface operation function temporarily, and the parameter settings are invalid.

4.9 More

Click **More** on the homepage interface of the system, and enter the customer password according to the prompts, you can view the control system version information and perform factory reset operations.

Figure 4-18 More Settings interface



The default client password is **0001**. If the user has not input the password before modifying the parameter value on the control system interface, the system will automatically pop up the password input box when clicking the input box of the corresponding parameter. Parameter value modification is allowed only after the correct password is entered.

5 Routine Maintenance

5.1 Overview

Routine maintenance is a kind of preventive maintenance, which means that maintenance personnel periodically carry out inspection and maintenance work during the normal operation of equipment, and timely discover and eliminate the alarms and hidden troubles in the equipment.

The purpose of routine maintenance is to enable the equipment to run stably for a long time. Through routine maintenance:

- It can find the related alarms in the equipment operation in time, avoid further serious alarms and ensure the stable operation of the equipment.
- It can find out the hidden trouble in time, eliminate the trouble in its infancy, and avoid the economic loss and the decrease of customer satisfaction after the trouble happens.
- It can find the running trend of equipment, implement optimization pertinently, and improve the running efficiency of equipment.

5.2 Unit Appearance Maintenance

Routine maintenance of unit appearance is shown in **Table 5-1**.

Table 5-1 Maintenance of unit appearance

Maintenance Item	Maintenance Standard	Maintenance Cycle	Detection Mode	Processing Method
Unit appearance	The unit is clean and dust-free and free of dirt	3 months	Visual inspection	Use brush or cotton cloth to remove dust and dirt from the unit.



The above maintenance periods are only suggested values, and users can adjust them according to the actual plan.

5.3 Electrical Cables and Terminals Maintenance

Maintenance of electrical cables and terminals is shown in **Table 5-2**.

Table 5-2 Maintenance of electrical cables and terminals

Maintenance Item	Maintenance Standard	Maintenance Cycle	Detection Mode	Processing Method
Reliability of electrical cables and	No looseness of power cable	3 months	Visual inspection	Tighten loose cables with a screwdriver after power off.

Maintenance Item	Maintenance Standard	Maintenance Cycle	Detection Mode	Processing Method
terminals of wiring panel	There is no looseness in the power terminal	3 months	Visual inspection	Re-fix the loose power terminal with a screwdriver after power off.
	There is no aging, damage, abnormal heating and other abnormalities in the power cable	3 months	Visual inspection	Replace the power cable after power off.
	There is no dust at the wiring panel	3 months	Visual inspection	Clean up the dust with a brush after power off.

NOTE

The above maintenance periods are only suggested values, and users can adjust them according to the actual plan.

5.4 Fans Maintenance

Routine maintenance of fans is shown in **Table 5-3**.

Table 5-3 Maintenance of indoor and outdoor fans

Maintenance Item	Maintenance Standard	Maintenance Cycle	Detection Mode	Processing Method
Operational reliability of fan	There is no dust in the fan and no foreign matter blockage at the tuyere	1 month	Visual inspection	After the power is off, clean the dust of the fan with a brush and clean the foreign matter at the tuyere.
	No abnormal noise in fan operation	1 month	Visual inspection	Use a screwdriver to tighten the fan after power off.

NOTE

The above maintenance periods are only suggested values, and users can adjust them according to the actual plan.

5.5 Drain Hole and Heat Exchangers Maintenance

CAUTION

The edges of the condenser and evaporator blades are sharp, please wear gloves for operation.

Routine maintenance of drain hole and heat exchangers is shown in **Table 5-4**.

Table 5-4 Maintenance of drain hole and heat exchangers

Maintenance Item	Maintenance Standard	Maintenance Cycle	Detection Mode	Processing Method
Condensate drain	The condensate drain is not blocked.	6 months	Visual inspection	Clear the blockage after power off.
Condenser cleaning	The condenser is not dirty and blocked.	6 months	Visual inspection	After the power is off, use a water gun to flush the condenser until the water after flushing becomes clear.
	No serious deformation of the fins	6 months	Visual inspection	Use tools such as tweezers or fin comb to calibrate after power off.
Evaporator cleaning	The evaporator is not dirty and blocked.	12 months	Visual inspection	Use a dry cotton cloth to wipe off the dirty blockage after power off.
	No serious deformation of the fins	12 months	Visual inspection	Use tools such as tweezers or fin comb to calibrate after power off.

NOTE

The above maintenance periods are only suggested values, and users can adjust them according to the actual plan.

6 Fault Management

6.1 Troubleshooting

6.1.1 Fans Fault

Common faults and recovery measures of fans are shown in **Table 6-1**.

Table 6-1 Fans fault

Fault Phenomenon	Possible Reasons	Inspection and Repair
The indoor fan does not run	The unit is not powered on	• Check if there is current at the unit power input terminal. • Check whether the external circuit breaker of the unit is switched on.
	The input power is abnormal, such as: power supply overvoltage, undervoltage, or lack of phase	Check whether the unit has corresponding alarms through RS485 communication.
	The unit is on standby status	Normal status. After the unit is powered on, it waits for 30s, and then enters the automatic control logic.
	The fan is stuck	Check if there are any internal components obstructing the operation of the fan, or foreign objects in the fan.
	The wiring terminal is loose	Tighten the terminals.
	The fan is damaged	If there is an internal circulation fan failure alarm, and the power connection is normal, and the fan has no foreign matter stuck, the fan may be damaged and the fan needs to be replaced.
	The control board is faulty	If the above possible causes are eliminated, check whether the main control board is faulty. If the main control board fails, replace it.

Fault Phenomenon	Possible Reasons	Inspection and Repair
The outdoor fan does not run	The compressor does not start	The external circulation fan starts only after the compressor is started, and the high-pressure side refrigerant needs to reach a certain pressure.
	The fan is stuck	Remove foreign objects.
	The wiring terminal is loose	Tighten the terminals.
	The fan is damaged	If there is no abnormality in the power connection of the condenser fan and the fan is not stuck with foreign matter, the fan may be damaged and the fan needs to be replaced.
	The control board is faulty	If the above possible causes are eliminated, please check whether the main control board is faulty. If the main control board fails, replace it.
Abnormal sound of fans	Fan bearing wear.	Replace the fan.
	Fan blades scratch other objects.	Check whether there are cables or other foreign objects interfering with the fan blades.
	Loose fan fixing bolts	Re-tighten the fixing bolts of the fan.

6.1.2 Cooling System Fault

Common faults and maintenance methods of cooling system are shown in **Table 6-2**.

Table 6-2 Cooling system fault

Fault Phenomenon	Possible Reasons	Inspection and Repair
Compressor not started	Unpowered (standby)	Check the main power switch, and check whether the unit has been turned on through the control system operation interface.
	Loose circuit connection	Tighten the circuit connector.
	Open or short circuit	Check the circuit for open circuit or short circuit, and repair the main power supply.

Fault Phenomenon	Possible Reasons	Inspection and Repair
	Inverter fault	Replace the inverter.
	Control board fault	Replace the control board.
	Compressor motor fault	Replace the compressor.
Compressor does not work	The control temperature is set too high and there is no cooling demand	Check the air inlet and outlet temperature, and set the appropriate control temperature.
High compressor discharge pressure	Dirty condenser	Clean the condenser.
	The external circulation fan does not run	Refer to **Table 6-1** to confirm the treatment measures for the phenomenon that the external circulation fan does not run.
	The internal circulation fan does not run	Refer to **Table 6-1** to confirm the treatment measures for the phenomenon that the internal circulation fan does not run.
Evaporator freezes	The internal circulation fan does not run	Refer to **Table 6-1** to confirm the treatment measures for the phenomenon that the internal circulation fan does not run.

6.2 Parts Replacement

6.2.1 Replace the Indoor Fans

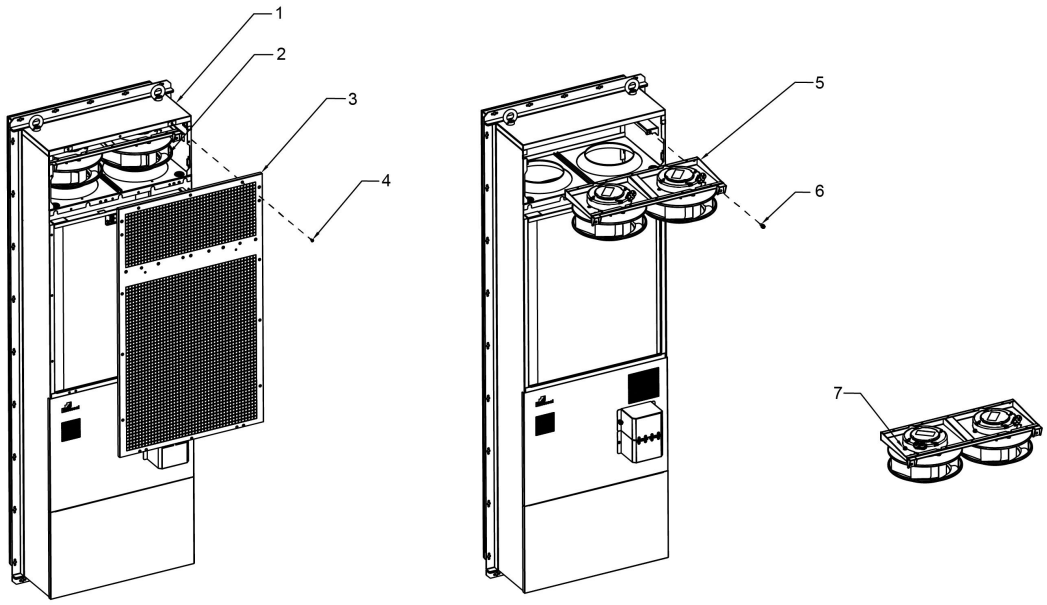
Prerequisites

- Power has been cut off.
- The tools flat-head screwdriver, Phillips screwdriver and diagonal pliers have been prepared.

Context

Operation instructions for replacing the indoor fans are shown in **Figure 6-1**.

Figure 6-1 Replacing the indoor fans



(1) Air conditioner

(2) Indoor fan

(3) Front cover

(4) Front cover screws (M4*10)

(5) Indoor fan fixing plate

(6) Indoor fan faxing plate screws

(7) Indoor fan installation screws (M6*16)

Procedure

Step 1 Use a Phillips screwdriver to remove the screws of the inner circulation panel (marked **3** in **Figure 6-1**, the screws are **M4*10** cross countersunk head screws).

Step 2 Use diagonal pliers to remove the cable tie of the indoor fan cable.

Step 3 Remove the wiring terminal of the indoor fan.

Step 4 Use a Phillips screwdriver to remove the screws of the indoor fan fixing plate (marked **6** in **Figure 6-1**, the screws are **M6*16** cross combination screws).

Step 5 Pull out the indoor fan and the indoor fan fixing plate.

Step 6 Use a Phillips screwdriver to remove the screws between the indoor fan and the fan fixing plate (marked **7** in **Figure 6-1**, the screws are **M6*16** cross combination screws).

Step 7 After the maintenance is completed, install the indoor fan in reverse order.

6.2.2 Replace the Outdoor Fan

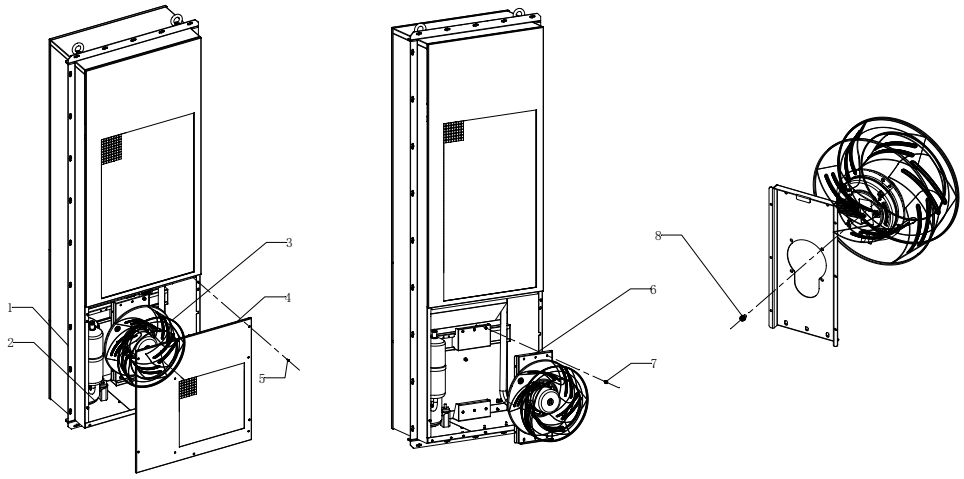
Prerequisites

- Power has been cut off.
- The tools flat-head screwdriver, Phillips screwdriver, special anti-theft screwdriver and diagonal pliers have been prepared.

Context

Operation instructions for replacing the outdoor fan are shown in **Figure 6-2**.

Figure 6-2 Replacing the outdoor fan



(1) Air conditioner	(2) Compressor	(3) Outdoor fan
(4) Outdoor fan maintenance panel	(5) Outdoor fan maintenance panel screws (M4*16 anti-theft screw)	(6) Outdoor fan fixing plate
(7) Mounting screws of the outdoor fan fixing plate (M6*16)	(8) Outdoor fan fixing screw (M6*16)	

Procedure

Step 1 Use a special anti-theft screwdriver to remove the maintenance panel of the outdoor fan (marked **4** in **Figure 6-2**, the screws are **M4*16** anti-theft screws).

Step 2 Remove the fixing tie of the outdoor fan cable.

Step 3 Remove the wiring terminal of the outdoor fan.

Step 4 Use a Phillips screwdriver to remove the screws of the fixing plate of the outdoor fan (marked **7** in **Figure 6-2**, the screw specification is **M6*16**).

Step 5 Use a Phillips screwdriver to remove the fixing plate of the outdoor fan, take out the outdoor fan and replace it (marked **8** in **Figure 6-2**, the screw specification is **M6*16**).

Step 6 After the maintenance is completed, install the outdoor fan in reverse order.

6.2.3 Replace the Compressor

CAUTION

- Please contact the manufacturer for after-sales treatment when replacing the compressor. It is strictly forbidden to replace it without permission.
- It is strictly forbidden to treat the failed compressor as unclassified municipal waste, and it must be treated independently according to relevant requirements.

6.2.4 Replace the Control Board

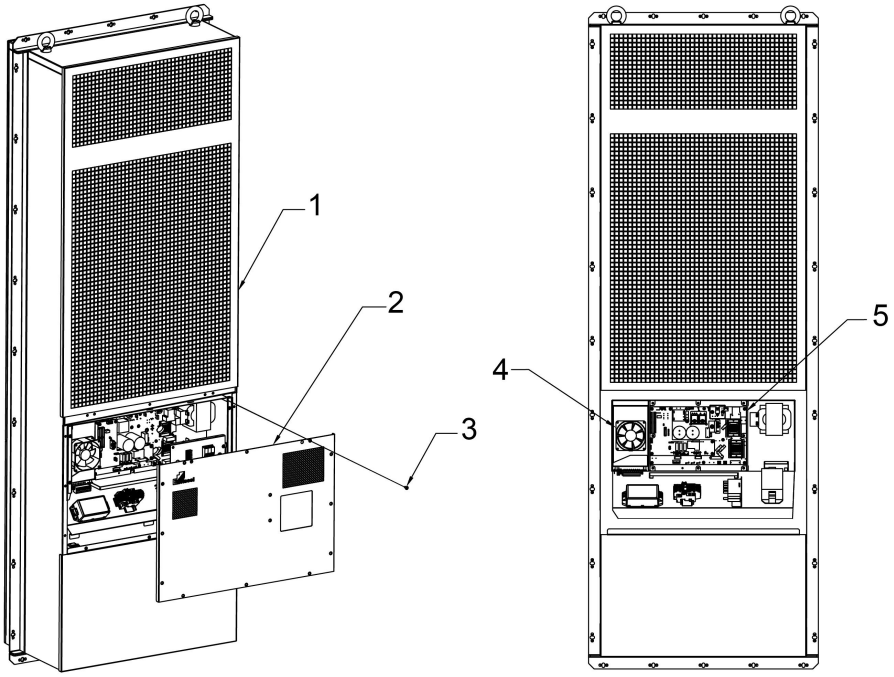
Prerequisites

- Power has been cut off.
- The tools flat-head screwdriver, Phillips screwdriver and clamp ammeter have been prepared.

Context

Operation instructions for replacing the control board is shown in **Figure 6-3**.

Figure 6-3 Replacing the control board



(1) Air conditioner (2) Electric control panel (3) Electric control panel screws (M4*10)

(4) Cooling fan (5) Drive and control integrated board

Procedure

Step 1 Use a Phillips screwdriver to remove the maintenance panel of the unit (marked **3** in **Figure 6-3**, the screw specification is **M4*10**).

Step 2 Remove all wiring terminals on the drive and control integrated board.

Step 3 Use a Phillips screwdriver to remove the fixing screws of the drive and control integrated board (the screws are **M4*16** combined screws).

Step 4 Take out the control board and replace it.

Step 5 After the maintenance is completed, install in the reverse order.

For the connection of electrical terminals, please refer to the internal wiring diagram of the air conditioner.

NOTE

- If the cables are damaged, please replace them at the same time.
- The steps for replacing other electronic control components are similar to those for replacing the control board.

6.3 Alarm Handling

The alarm information of the unit is shown in **Table 6-3**.

Table 6-3 Alarm list

Alarm Item	Generation Conditions	Default processing	Troubleshooting	Solutions
Return air high temperature alarm	The return air temperature is higher than the alarm setting value	Unit keeps running in cooling mode	1. Check whether the high temperature alarm setting value is reasonable. 2. Check whether the cooling capacity of the unit meets the demand.	1. Restore the factory settings of the system, or call the service hotline to confirm the reasonable value and then set it. 2. Increase the number of air conditioners.
Return air low temperature alarm	The return air temperature is lower than the alarm setting value	Unit keeps running in heating mode	1. Check whether the low temperature alarm setting value is reasonable. 2. Check whether the electric heater is faulty.	1. Restore the factory settings of the system, or call the service hotline to confirm the reasonable value and then set it. 2. Replace the electric heater when it is faulty.

Alarm Item	Generation Conditions	Default processing	Troubleshooting	Solutions
Return air high humidity alarm	The return air humidity is higher than the alarm set value	Unit keeps running in dehumidification mode	1. Check whether the return air humidity set point or high humidity alarm set point is reasonable. 2. Check if any doors and windows are open to cause air circulation inside and outside.	1. Restore the factory settings of the system, or call the service hotline to confirm the reasonable value and then set it. 2. Close the doors and windows and check whether the sealing measures between the unit and the installation surface are normal.
Return air temperature sensor failure alarm	The communication between the return air temperature sensor and the control board is interrupted or the sensor is damaged	The unit performs logical control according to the temperature sense of the outlet air	1. Check whether the electrical control wiring of the return air temperature sensor and whether the cable is damaged. 2. Check whether the return air temperature sensor is faulty. 3. Check whether the control board is faulty.	1. Fasten the connection terminal or replace the cable. 2. Replace the sensor when it is faulty. 3. Replace the control board when it is faulty.
Internal circulation fan failure alarm	The communication between the internal fan and the control board is interrupted or the fan is damaged	The unit stops running	1. Check whether the connecting terminal is loose and whether the cable is damaged. 2. Check if there is any foreign matter stuck to the fan. 3. Check whether the internal fan is faulty. 4. Check whether the control board is faulty.	1. Fasten the connection terminal or replace the cable. 2. Clean up foreign objects. 3. Replace the fan when it is faulty. 4. Replace the control board when it is faulty.

Alarm Item	Generation Conditions	Default processing	Troubleshooting	Solutions
System high pressure alarm	The system pressure reaches the high pressure alarm trigger point	The unit stops running	1. Check whether the ambient temperature is too high. 2. Check whether the wiring of the high-pressure switch is loose, or the cable is damaged causing a false alarm. 3. Check whether the condenser is severely dirty and blocked. 4. Check whether there is debris blocking the air inlet. 5. Check whether the condenser fan is out of power or working abnormally.	1. Place the unit correctly according to the installation requirements and avoid exposing the unit to direct sunlight. 2. Fasten the wiring of the high-pressure switch or replace the cable. 3. Clean the condenser. 4. Clean up the dirty blockage in front of the return air inlet of the unit. 5. Replace the condenser fan when it is not due to power failure.
Frequent high pressure alarm locking	3 times of system high pressure alarm triggered within 1 hour	The unit stops running	1. Check whether the ambient temperature is too high. 2. Check whether the wiring of the high-pressure switch is loose, or the cable is damaged causing a false alarm. 3. Check whether the condenser is severely dirty and blocked. 4. Check whether there is debris blocking the air inlet. 5. Check whether the condenser fan is out of power or working abnormally.	1. Place the unit correctly according to the installation requirements and avoid exposing the unit to direct sunlight. 2. Fasten the wiring of the high-pressure switch or replace the cable. 3. Clean the condenser. 4. Clean up the dirty blockage in front of the return air inlet of the unit. 5. Replace the condenser fan when it is not due to power failure.

Alarm Item	Generation Conditions	Default processing	Troubleshooting	Solutions
System low pressure alarm	The system pressure reaches the low pressure alarm trigger point	The unit stops running	1. Check for refrigerant leakage. 2. Check whether the electronic expansion valve is faulty.	1. Check for leaks and replenish refrigerant after repairs. 2. Replace the electronic expansion valve when it is faulty.
Frequent low pressure alarm locking	3 times of system low pressure alarm triggered within 1 hour	The unit stops running	1. Check for refrigerant leakage. 2. Check whether the electronic expansion valve is faulty.	1. Check for leaks and replenish refrigerant after repairs. 2. Replace the electronic expansion valve when it is faulty.
AC over-voltage alarm	The input AC voltage is higher than the operating voltage range	The unit stops running	1. Check the power supply voltage of the unit. 2. Check whether the lightning protection component on the driver board is faulty.	1. Check whether the power supply grid is stable. 2. Replace the lightning protection component when it is faulty.
AC under-voltage alarm	The input AC voltage is lower than the operating voltage range	The unit stops running	1. Check the power supply voltage of the unit. 2. Check whether the lightning protection component on the driver board is faulty.	1. Check whether the power supply grid is stable. 2. Replace the lightning protection component when it is faulty.
Abnormal power down alarm	The AC input is disconnected in the power-on state	Unit power off and shut down	The power supply grid is cut off.	Check the power supply grid.

Alarm Item	Generation Conditions	Default processing	Troubleshooting	Solutions
Suction temperature sensor failure alarm	The sensor communication is interrupted or the sensor is damaged	The unit stops running	1. Check whether the electrical control wiring of the suction temperature sensor is loose and whether the cable is damaged. 2. Check whether the suction temperature sensor is faulty. 3. Check whether the control board is faulty.	1. Fasten the electrical control wiring or replace the cable. 2. Replace the suction temperature sensor when it is faulty. 3. Replace the control board when it is faulty.
Evaporator freezing alarm	The evaporator coil temperature is lower than -2℃	The unit stops running	1. Check whether the filter is dirty and blocked, resulting in a small air volume. 2. Check whether the evaporator is dirty or blocked and affects heat dissipation.	1. Clean the filter. 2. Clean the evaporator.

7 Reference

Figure 7-1 shows a barcode example for an air conditioner.

Figure 7-1 Air conditioner barcode sample



- ①: Customized by the manufacturer (M01XXXX***XXXX).
- ②: Production date, for example, 21621 means June 21, 2021. A/B/C means October/November/December.
- ③: The serial number of this batch of products.

前言

概概述

本文档为 MC90HDNC1L/R 系列机柜空调的通用版本说明书，主要介绍该系列空调器的产品概述、安装、控制器操作、例行维护和故障处理，方便读者了解该系列空调的使用和维护。

读者对象

本文档主要适用于以下读者：

- 产品安装工程师
- 技术支持工程师
- 售后维护工程师

符号约定

文中可能出现以下标志，代表的含义如下。

符号	说明
	警示存在严重危险的场景，若不遵守要求，可能导致死亡或严重的人身伤害。
	警示存在潜在危险的场景，若不遵守要求，可能造成中度或轻微的人身伤害。
	警示影响设备安全或环境安全的场景，若不遵守要求，可能对设备等造成破坏或其他不良结果。
	提示一般性细节，或其它有用的、可以简化工作的辅助信息。

修订记录

修改记录累积了每次文档更新的说明。最新版本的文档包含以前所有文档版本的更新内容。

版本	更新	修改日期
A1	首次发布。	2024-06-14

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1 安全注意事项

请仔细阅读本章后再进行装配、安装、电气连接和维护作业，并在执行相关操作时严格遵守相关要求，谨慎操作。

声明

以下情况本公司不承担责任：

- 在不满足该手册描述应用场景的异常条件或环境中运行设备。
- 未遵守本手册的操作要求，因不当用途，或不正确的安装、维护方式等导致设备故障。
- 非本设备造成的故障（如由用户的设备、软件等造成的故障）。
- 用户自行更换或拆装产品零部件造成的设备损坏，或由非授权维修服务者拆修而造成的设备损坏。
- 由于火灾、地震、洪水等不可抗力而造成的设备损坏。

通用安全

- 误操作导致任何人身伤害都应该在第一时间处理，务必立即获得急救或医疗护理。
- 请严格遵守所在地的安全规范和法律法规，避免意外事故的发生。手册中的安全注意事项不代表应该遵守的所有安全事项，仅作为当地安全规范的补充。
- 若设备上粘贴有警示标签，操作前请务必认真阅读并严格遵守。
- 仅允许使用原厂部件，以及经过制造商授权、用于确保空调设备稳定性和安全性的部件。如果使用其他部件则保修权利失效。
- 不允许在未与制造商协定并经过其同意的情况下，对空调设备进行任何更改。
- 不允许认知和协调能力受限的人员和儿童操作、维护、清洁设备或将其用作玩具。
- 进行安装维护作业请委托专业技术人员，自行操作可能导致漏水、触电或火灾等事故发生。

说明

专业人员指具备专业培训、经验和技能,以及熟悉事故防范和操作要求相关的条例、规定和措施的人员，并且此人具备发现并规避所有潜在危险的能力。由运营方或负责人授权该人员确保设备的安全性、进行所有必要的操作。

电气安全

- 禁止带电安装、拆除电源线。安装、拆除电源线前请关闭电源开关。
- 安装设备时，必须先接地。拆除设备时，最后再拆地线。禁止在未安装接地导体时对设备上电。
- 请在断电后进行安装维护操作，并且在操作过程中请勿接通电源。某些需要运行时进行的操作（如测量电流、压力、温度等），必须确保设备接线已经全部正确完成，才可接通电源，操作完成后应立即切断电源。

警告

机组中包含的电容在断电后仍会持续放电，为了确保人身安全，请在机组断电后至少 1 分钟以上再进行维护操作。

- 进行电气维护时必须做防护措施，如佩戴绝缘手套或穿绝缘鞋等。

操作安全

- 本设备不适用于身体、感官或精神能力下降或缺乏经验和知识的人员（包括儿童），除非负责其安全的人员对其使用本设备进行了监督或指导。
- 应监督儿童，确保他们不玩电器。
- 电器的安装应符合国家接线规定。
- 机组运行时请勿用手随意触碰机组内部器件，避免触电或被风机扇叶所伤。
- 机组运行或调试过程中如果出现剧烈振动或异常声响，请停止一切操作并立即切断电路开关进行检查。
- 检修维护时应断开全部的电路开关，并设立警戒防止他人误闭合电路开关。
- 仅当断开全部的电路开关，且内部控制板不再闪烁信号灯时，才可以对设备线路和电子器件进行操作，且必须佩戴防静电手套。
- 安装维护过程中请勿让水等液体进入设备的接线端子区域。
- 禁止独自一人手工搬运较重的设备，手工搬运较轻设备时请佩戴保护手套。
- 操作部件存在利物（如翅片等）时，请佩戴保护手套。
- 剪断扎线带时请小心，避免割伤线缆。

应用安全

- 请勿在易燃易爆环境中使用本产品。
- 该系列机柜空调为户外应用型空调，若需将空调应用于室内，请与产品生产厂家进行确认后实施必要的防水措施。
- 涉及专业的维护时请谨慎操作，具体请咨询产品生产厂家的售后客户服务中心。

回收处理



严禁将该产品作为未分类的城市垃圾进行报废处理，请进行分类回收利用。报废处理必须遵循相关要求，采取特殊方式进行独立处理，避免对环境造成潜在污染。

2 产品描述

2.1 特性

2.1.1 产品简介

型号说明

MC 系列一体式空调的命名规则如图 2-1 所示。

表 2-1 命名规则

MC	90	H	D	N	C	1	L/R
1	2	3	4	5	6	7	8

1 产品系列	MC：储能一体式空调系列
2 额定制冷量	90：9kW
3 基本功能	H：含加热
4 安装方式	D：嵌入式门装
5 应用环境	N：常温
6 噪音等级	C：普通噪音
7 输入电源	1：单相交流输入
8 版本代码	L 或 R

特性介绍

MC90HDNC1L/R 机柜空调是针对机柜或集装箱等应用环境而开发的一款制冷产品，适用于以下场景：

- 机柜等内部设备发热量大。
- 机柜等内部设备对环境温度敏感。
- 机柜等内部设备需要与外部完全隔离。

MC90HDNC1L/R 机柜空调主要具备以下特性：

表 2-2 MC90HDNC1L/R 空调特性

特性	说明
设计先进	• 采用一体式结构，安装简单便捷，通电后即可工作，无需复杂调测。 • 采用户外金属构造，配合防腐措施。
功能齐备	• 支持智能制冷、加热和除湿功能。 • 支持 RS485 Modbus 通讯功能。 • 支持断电记忆和自启动功能。
性能稳定	可保持 24 小时×365 天持续运行。
高效节能	采用交流供电，配备变频压缩机和 EC 风机。

2.1.2 功能描述

智能制冷/制热/除湿

机组配置高效控制器，并提供显示器用于查看机组状态和设置参数，操作方便。

控制器实时监测柜内回风温湿度，并根据目标温湿度设定进行智能调节，通过控制压缩机、风机、电加热器等部件的运行，实现智能制冷、制热和除湿功能。

RS485 Modbus 通讯

机组对外提供 RS485 通信接口，通过 Modbus 协议与上位机进行通讯。上位机通过发送请求消息，可进行机组各部件运行状态查询和控制参数设置等操作。

断电记忆和自启动

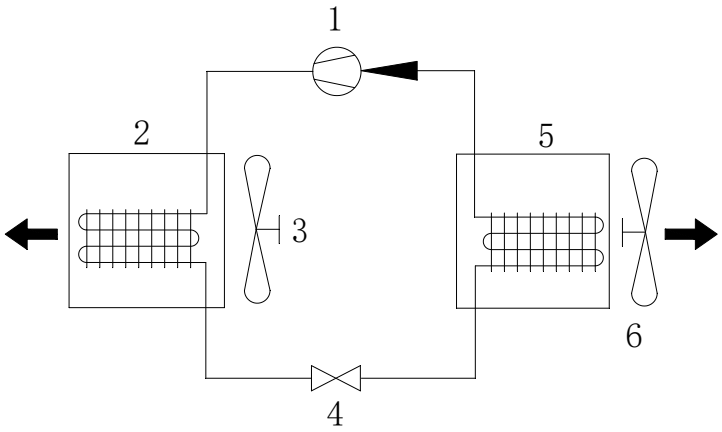
机组在运行过程中掉电时会记忆掉电之前的参数设置信息，再次来电后自行启动，并根据断电前的参数设置自动智能运行。

2.2 实现

2.2.1 系统原理

MC 系列一体式空调主要包括压缩机、节流元件、室内外风机、蒸发器、冷凝器和电控盒，系统实现原理如图 2-1 所示。

图 2-1 系统实现原理图



1：压缩机　　2：冷凝器　　3：外循环风机

4：节流元件　　5：蒸发器　　6：内循环风机

- 压缩机：负责对制冷剂进行压缩，为制冷系统提供动力。
- 冷凝器：采用翅片管式设计，换热效率高。制冷剂在冷凝器中冷凝放热。
- 节流元件：采用电子膨胀阀，通过调节开度对制冷剂流量进行控制。
- 蒸发器：采用翅片管式设计，换热效率高。制冷剂在蒸发器中蒸发吸热。
- 内风机：负责将室内空气吹过蒸发器盘管表面进行热交换。
- 外风机：负责将冷凝器盘管里制冷剂冷凝时释放的热量排出到室外。
- 控制盒：对空调部件进行逻辑控制，实现温湿度控制需求。

2.2.2 工作流程

MC 系列一体式空调的工作流程如下：

1. 当机柜或集装箱等的室内温度达到制冷开启点后压缩机启动，压缩机对制冷剂进行压缩。

   制冷剂经过压缩机压缩后转变为高温高压的气体进入冷凝器。

2. 冷凝器将气态的制冷剂冷凝为液态。

   外风机带走制冷剂冷凝时释放的热量，制冷剂进入节流元件。

3. 节流元件对制冷剂进行降压降温。
4. 低温低压的湿蒸汽制冷剂进入蒸发器吸收热量。

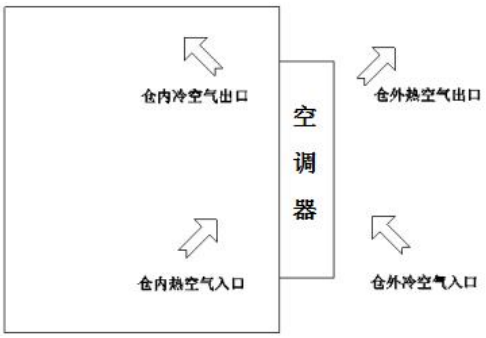
   内风机带走内部空间热量，达到制冷效果。

5. 湿蒸汽制冷剂经过蒸发器吸热后，变成低温低压的气体进入压缩机进行压缩，循环进行制冷过程。

2.2.3 风路设计

MC 系列一体式机柜空调的风路循环系统如图 2-2 所示。

图 2-2 气流循环示意图



- 内循环风路中，空调器从下部吸入柜内热空气，经过热交换后从上部排出冷空气。
- 外循环风路中，空调器从下部吸入室外冷空气，经过热交换后从上部排出热空气。

2.2.4 控制逻辑

MC90HDNC1L/R 一体式机柜空调的控制逻辑如表 2-3 所示。

表 2-3 温度控制逻辑

项目	控制逻辑	说明
制冷	• 制冷开启条件：制冷需求 ≥ 50% • 制冷停止条件：制冷需求 ≤ 0%	制冷需求=（室内回风温度-制冷设定点【23°C】/制冷回差【5°C】* 100%
加热	• 加热开启条件：加热需求 ≥ 100% • 加热停止条件：加热需求 < 0%	加热需求=（制热设定点【15°C】-室内回风温度）/制热回差【5°C】 * 100%
除湿	• 除湿开启条件：除湿需求 ≥ 100% • 除湿停止条件：除湿需求 < 0%	除湿需求(%) =（室内回风湿度 - 除湿设定点【默认 60%】）/除湿灵敏度【默认 15%】*100%

表 2-4 部件工作逻辑

部件	工作逻辑
压缩机	机组默认的制冷设定点为 23°C，回差为 5°C。缺省条件下制冷启动温度为 25.5°C（23°C+5°C*50%=25.5°C），制冷停止温度为 23°C（23°C+5°C*0%=23°C）。 机组默认的除湿设定点为 60%，回差为 15%。缺省条件下除湿开启湿度为 75%（60%+15%*100%=75%），除湿停止湿度为 60%（60%+15%*0%=60%）。 • 当室内温度高于 25.5°C 时，压缩机启动运行。 • 当室内温度低于 23°C 时，压缩机停止运行。 • 当室内回风湿度高于除湿开启点 75%时，压缩机启动运行。 • 压缩机每次停机需等待三分钟后才能再次启动，避免频繁启停导致压缩机故障。

部件	工作逻辑
内风机	空调开启后内风机工作模式如下： • 送风模式：室内温度小于制冷设定点时，空调工作于送风模式，内风机保持在较低速度运行。 • 制冷模式：内风机以高速运行。 • 除湿模式：内风机以低速运行。
外风机	• 外风机根据冷凝压力来调节其转速。 • 当冷凝压力较低时，外风机通过变速控制降低转速，实现降低噪声的目的。
电加热	机组默认的制热设定点为 15°C，回差为 5°C。缺省条件下加热启动温度为 10°C（15°C-5°C*100%=10°C），加热停止温度为 15°C（15°C-5°C*0%=15°C）。 • 当室内回风温度低于 10°C 时，电加热启动运行。 • 当室内回风温度高于 15°C 时，电加热停止运行。

2.3 规格

空调相关技术参数规格见表 2-5。

表 2-5 参数规格

参数	规格
运行电压范围	187VAC ~ 253VAC 50/60Hz
运行环境温度范围	-30°C ~ 55°C
运行相对湿度范围	5% ~ 95%
贮存环境温度范围	-40°C ~ 75°C
贮存环境湿度范围	5% ~ 95%
运输性能	能适应陆运、空运和海运等方式运输
外形尺寸(W×D×H)	1770mm×620mm×300mm

说明

设备铭牌上标注的技术指标此处不重复描述。

3 安装指南

3.1 安装须知

3.1.1 工具准备

进行空调的安装维护前请准备不限于表 3-1 中列举的常用工具，现场技术人员可根据实际情况酌情增减。

表 3-1 安装工具

工具	图示	功能
一字螺丝刀（2mm×75mm）		一字螺丝刀安装或拆除控制器等的连接线。
十字螺丝刀（M3/M4/M6/M8）		十字螺丝刀安装或拆卸部件的螺钉。
斜口钳		现场加工线缆，去除固线扎带。
钳型电流表		• 现场检测工作电压、电流。 • 现场检测内部配电电压、电流。
卷尺		现场测量尺寸。
压线钳		现场加工线缆，压接各式端子。
冲洗工具		使用干燥压缩空气清洁空调系统冷凝器。
毛刷		清理灰尘。

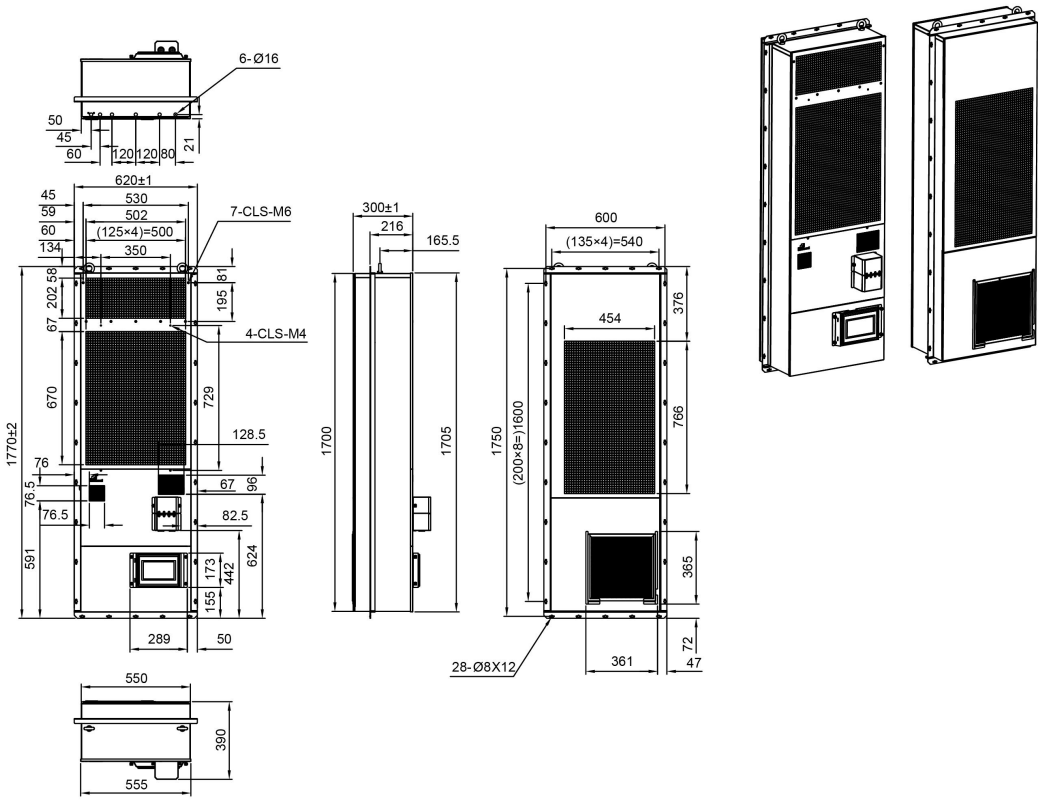
3.1.2 运输与搬运

- 在运输和搬运过程中，请确保外包装完整，避免受潮或损坏外包装，并确保机组免受碰撞、雨淋和化学腐蚀。
- 拆开包装之前，尽量将空调搬运到距离安装地点最近的地方。
- 在搬运和放置时，必须保持空调正立。

3.1.3 设备外形尺寸

MC90HDNC1L/R 机柜空调的外形尺寸如图 3-1 所示，供安装过程中参考。

图 3-1 外形尺寸

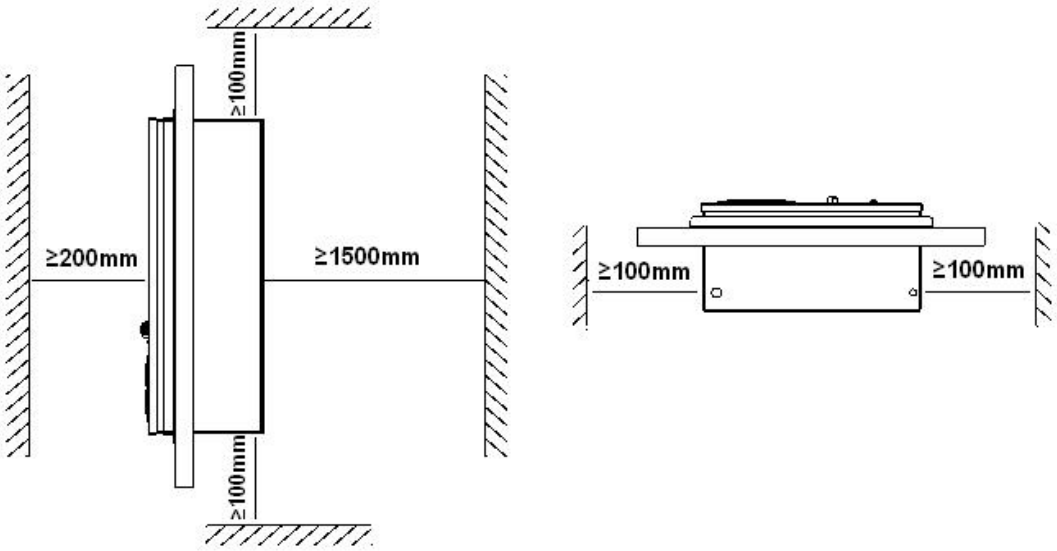


3.1.4 安装布局

为确保机组的正常运行，内/外循环风口处不允许放置物品，阻碍空气流动。

请参考图 3-2 将空调安装在固定装置（如机柜门）上。

图 3-2 安装布局参考图



机组安装维护空间的具体要求如下所示：

表 3-2 安装维护空间要求

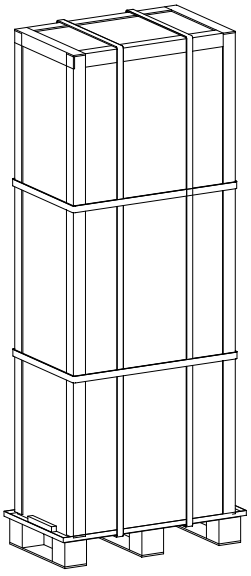
项目	要求
室内维护空间	内循环风机的进出风口前至少 200mm 处避免安装或摆放任何设备，否则可能导致风路阻碍。
室外维护空间	外循环风机的进出风口前至少 1500mm 处避免安装或摆放任何设备，否则可能导致风路阻碍。

3.2 开箱与验收

请尽量将机组搬运到离安装位置最近的地点再进行开箱拆卸，以方便移动机组。同时确保机组正立放置，不允许平放或者侧卧。

本产品采用纸箱包装，拆除包装时请先将包装外部的打包带拆除，然后打开纸箱，如图 3-3 所示。

图 3-3 拆除包装示意图



如果不是立即安装产品，或需要转运产品至其他地区时，请在开箱检查完成后重新对机组进行包装。

开箱后请做好如下检查：

- 请仔细检查机组是否存在因运输造成的破坏，尤其关注是否存在部件松动、凹坑、擦伤和漏油等现象。如果发现了任何损坏，请在一周内连同包装外的条形码编号上报到销售商。
- 请按照附件清单清点机组和安装附件的数量。
- 处理包装材料前，请务必确认包装材料里未掺杂任何松动掉落的部件。为了保护环境，建议对包装箱进行回收利用。

3.3 机械安装

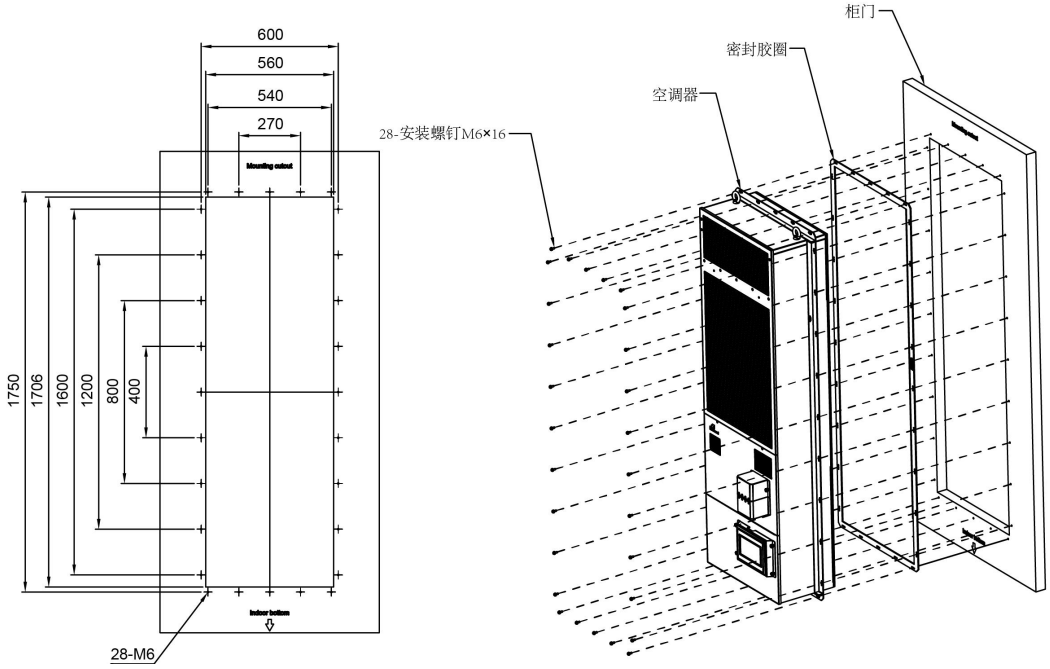
前提条件

- 已参考“3.1.3 设备外形尺寸”和“3.1.4 安装布局”合理规划机组的安装区域。
- 已准备好安装工具和所需附件。

背景信息

MC90HDNC1L/R 机组的安装如下图所示。

图 3-4 安装示意图



操作步骤

步骤 1 将机组移动至规划的安装位置。

步骤 2 使用十字螺丝刀和螺钉将空调紧固在机柜门上。

步骤 3 将冷凝水管连接至规划的排水管道。

3.4 电气连接

前提条件

已完成机械安装。

注意事项

警告

- 电气连接必须由受过培训的专业人员进行操作。
- 连接电源线时，禁止带电操作。
- 主电源必须满足机组要求，具体请以产品铭牌参数为准。
- 为了避免任何破坏性的影响，接通电源前必须将机组可靠接地。
- 所有电气连接操作都必须符合国家和当地电气规范要求。

操作步骤

步骤 1 连接电源线。

MC90HDNC1L/R 机柜空调的电源接口如图 3-4 所示，接口定义如表 3-3 所示。

图 3-4 电源接口

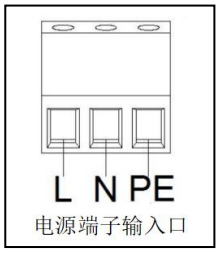


表 3-3 电源接口定义

脚位	定义	描述
L	单相电源 L（L1）	建议使用 3×10mm^2（3×8AWG）规格的线缆，仅允许使用铜线。
N	单相电源 N（L2）	
PE	单相电源 PE	

步骤 2 连接通讯线。

MC90HDNC1L/R 机柜空调的通讯接口如图 3-5 所示，接口定义如表 3-4 所示。

图 3-5 通讯接口

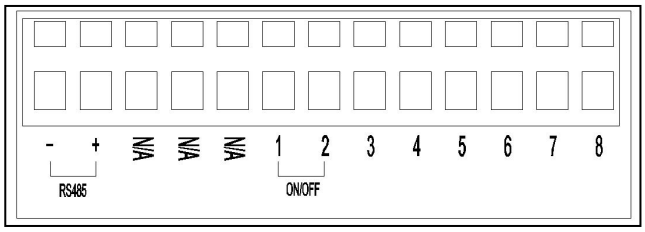


表 3-4 通讯接口定义

类型	对应脚位	脚位定义	说明
通讯接口	RS485-	RS485-	上位机通讯接口，建议采用 2×0.5mm² 双绞屏蔽线。
	RS485+	RS485+	
	N/A	-	-
	N/A	-	
	N/A	-	
	1	远程开关机干接点-	建议采用 2×0.5mm² 双绞屏蔽线。
	2	远程开关机干接点+	
	3	-	-
	4	-	
	5	-	
	6	-	
	7	-	
	8	-	

步骤 3 将 5PIN 显示屏端子排对插连接到整机。

步骤 4 连接完成后用钳形电流表检查电源线和信号线是否存在短路、断路的现象。

3.5 安装后检查

完成安装后请参考表 3-5 中列举信息进行再次检查，仅当全部检查通过后才允许上电开机。

表 3-5 安装后检查

检查项	检查标准
机组安装	机组水平放置，紧固零件已锁紧。
	冷凝水排水管已接入排水管道。

检查项	检查标准
	设备内部或周围的杂物（如运输材料、结构材料和工具等）已经清除。
电气连接	电源电压与设备铭牌上的额定电压相同。
	所有电缆、电路接头已紧固，紧固螺钉无松动。
	电缆无破损等现象。
	系统电气回路没有开路、短路和接错等现象。

4 控制器

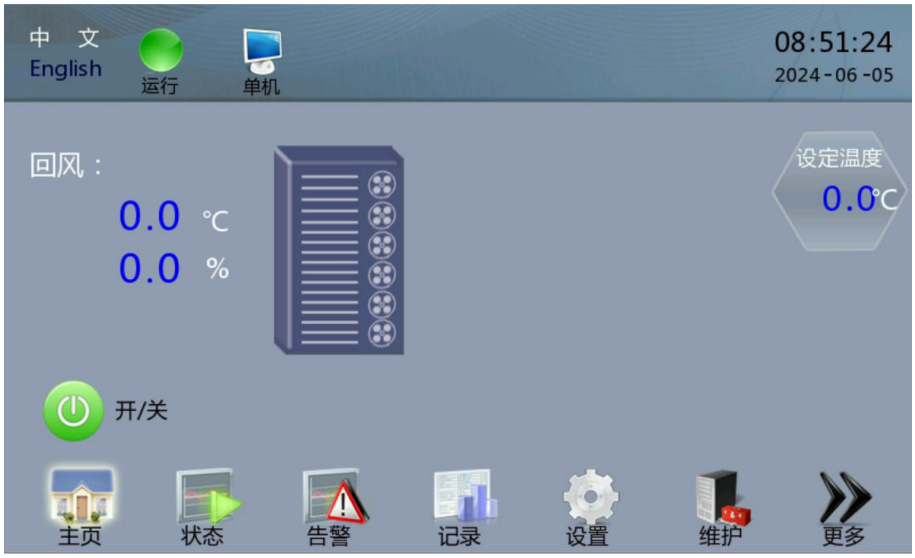
4.1 简介

MC90HDNC1L/R 机组空调支持外拉手操器，向用户提供人机交互界面。用户通过触摸屏可便捷地进行查询、设置和监控等操作，确保机组处于正常的运行状态。

4.2 控制系统界面

闭合电源开关对机组上电，系统初始化后进入控制系统主界面，如图 4-1 所示。

图 4-1 系统初始化界面

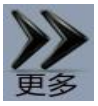


系统上电后，在控制系统界面按任意键，背光灯亮。持续一段时间后无键盘操作时，背光变暗。

机组控制界面中各个功能图标的含义如表 4-1 所示。

表 4-1 控制界面功能图标

图标	说明
	开关机按钮图标，单击后可改变机组当前的工作状态。
	显示语言图标，单击后可来回切换当前界面的语言为中文或英文。
	工作状态图标，以指示灯的形式显示当前机组的工作状态。 • 绿色表示运行状态。 • 红色表示关机状态。

图标	说明
	系统主页图标，单击后进入控制系统的操作主页界面，可查看机组的当前时间、运行模式、回风温度、湿度和界面语言。
	系统状态图标，单击后进入“系统状态”界面，可查看环境状态、运行状态及群控状态。
	系统告警图标，单击后进入“告警”界面，可查看当前告警及历史告警。
	系统运行记录图标，单击后进入“记录”界面，可查看机组的运行记录、运行日志和运行曲线。
	系统设置图标，单击后进入“参数设置”界面，可进行温湿度设置、群控设置、系统配置、账号设置和时间设置。
	系统维护图标，单击后进入“维护”界面，可进行健康模式状态，加湿排水的设置。
	系统更多功能图标，单击后进入“更多设置”界面，可进行客户密码输入，和查看控制系统版本信息。

4.3 系统开关机

注意

- 机组上电后，在压缩机不开启的情况下，加热带自动开启。当机组首次上电后，需确保在开机前，压缩机加热带的加热时长不低于 4 小时。
- 当机组长时间（约 4 个月）未启用时，必须在开机前为机组上电使压缩机加热带工作（循环水泵不要启动），将积留在压缩机内的制冷剂液体加热挥发，否则直接开机将会对压缩机产生不良影响。压缩机加热带的加热时长须不低于 10 小时。

每次单击“开/关”按钮后，控制系统切换机组的工作状态。

开机时系统弹出的对话框如图 4-2 所示。

图 4-2 系统开机



开机后，机组由关机状态更新为运行状态。在任何界面上，如果持续 180s 无键盘操作，控制系统自动返回主界面。

关机时系统弹出的对话框如图 4-3 所示。

图 4-3 系统关机



关机后，机组由运行状态更新为关机状态。

系统支持断电记忆功能，若系统断电后再上电，系统自动进入掉电前的状态（如：若断电前系统为开机状态，则上电后系统自动开机进入正常显示界面）。

4.4 状态

在控制系统主界面上单击“状态”，可查看机组当前的环境状态、运行状态和群控状态。

图 4-4“环境状态”界面



图 4-5“运行状态”界面



图 4-6“群控状态”界面



4.5 告警

在系统主界面上单击“告警”，可查看机组的当前告警和历史告警信息。

图 4-7“当前告警”界面



单击“选项 > 复位告警”，可对当前告警进行复位。

图 4-8“历史告警”界面



4.6 记录

在系统主界面上单击“记录”，可查看机组当前的运行记录、运行日志和运行曲线。

图 4-9“运行记录”界面



机组出厂时运行时间和动作次数初始值都为 0，其值根据实际运行情况自动累加。

图 4-10“运行日志”界面



图 4-11“曲线”界面



系统默认显示室内设定值（制冷设定值、除湿设定值）曲线、室内实际值（实际回风温度、实际湿度）曲线、室外实际值（室外实际温度）曲线。操作用户在“曲线”界面可进行以下操作：

- 通过去勾选曲线类型前的复选框，定制需要展示的曲线。
- 单击“选项 > 清除曲线”，清除当前界面展示的曲线。

  清除曲线后，控制系统根据机组运行情况自动重新生成曲线。
- 单击曲线展示区域四周的温度值和压力值，可根据实际规划修改温度值和湿度值范围。

4.7 设置

温湿度设定

注意

- 为使机组运行可靠并发挥最大能效，建议用户谨慎更改温度设定点。
- 当参数设定值不满足实际运行需求需进行修改时，请与产品售后进行确认。

在系统主界面上单击“设置”后选择“温湿度设定”页签，系统显示如图 4-12 所示，界面的参数设置如表 4-2 所示。

图 4-12“温湿度设置”界面



表 4-2“温度设置”界面参数说明

参数名	缺省值	设置范围	单位	说明
制冷点	23	[7~50]	°C	制冷、制热、除湿的开启和关闭条件请参见“表 2-3”。
制冷灵敏度	5	[1~15]	°C	
加热点	15	[-40~20]	°C	
加热灵敏度	5	[1~15]	°C	
除湿点	60	[10~90]	%	
除湿灵敏度	15	[3~20]	%	

参数名	缺省值	设置范围	单位	说明
温度控制方式	回风热点	• 回风平均 • 出风平均 • 回风热点 • 出风热点 • 监控下发	-	标识系统控制的温度类型。 • 回风平均：系统根据回风温度的平均值进行控制。 • 出风平均：系统根据出风温度的平均值进行控制。 • 回风热点：系统根据回风温度的最高值进行控制。 • 出风热点：系统根据出风温度的最高值进行控制。 • 监控下发：系统根据上位机通讯时下发的电芯温度进行控制。
湿度控制方式	• 回风平均	• 回风平均 • 出风平均 • 群控平均回风 • 群控平均出风 • 监控下发	-	标识系统控制的湿度类型。 • 回风平均：系统根据回风湿度的平均值进行控制。 • 出风平均：系统根据出风湿度的平均值进行控制。 • 群控平均回风：系统根据群控情况下多台机组回风湿度的平均值进行控制。 • 群控平均出风：系统根据群控情况下多台机组出风湿度的平均值进行控制。 • 监控下发：系统根据上位机通讯时下发的电芯湿度进行控制。

说明

- 以上参数缺省值和取值范围仅供参考，具体以出厂设置为准。
- 表中未列举参数保持出厂默认值。

系统配置

在系统主界面上单击“设置”后选择“系统设置”页签，系统显示如图 4-13 所示，界面的参数设置如表 4-3 所示。

图 4-13“系统配置”界面

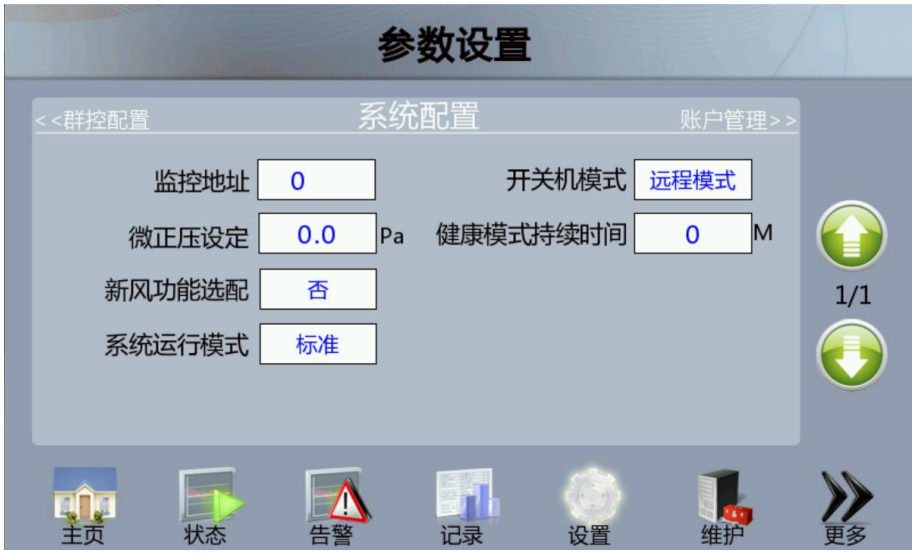


表 4-3“系统配置”界面参数说明

参数名	缺省值	设置范围	说明
监控地址	1	[1~247]	机组的监控地址，多个机组时取值保证唯一。
开关机模式	远程模式	• 本地模式 • 远程模式	机组支持按键开关机、监控开关机（Modbus 协议开关机）两种开关机方式。 • 本地模式：监控开关机无效。以按键关机进行关机时，机组进入关机状态。两种方式都为开机状态时，机组进入开启状态。 • 远程模式：以两种关机方式中任一方式关机时，机组进入关闭状态。两种方式都为开机状态时，机组进入开启状态。

说明

- 以上参数缺省值和取值范围仅供参考，具体以出厂设置为准。
- 表中未列举参数保持出厂设置的默认值。

群控配置

注意

配套群控功能时，操作员必须在机组首次上电正常运行后，进行相关的群控设置，否则将无法实现群控功能，机组返回至独立运行控制模式。

群组控制通过多个机组级联，可以提高机组的使用效率，避免机组间的竞争运行。

当群控为掉线状态时，配置界面的参数取值显示为“掉线”，如图 4-14 所示。

图 4-14“群控配置”界面



配置群控功能时，可参考表 4-4 进行参数设置。

表 4-4“群控配置”界面参数说明

参数名	说明
群控模式	配套群控功能时，设置为“群控”。
本机地址	本机的地址，主控机组的地址配置为“1”。 如群控机组数量为 2，即两台机组互相备份时，配置本机地址为“1”，另一台备份的机组地址配置为“2”。
主备模式	设置为“自动”。
成员数量	群组的数量。 如两台机组互相备份时，配置为“2”。
备机数量	备机的数量。 如两台机组互相备份时，配置为“1”。

参数名	说明
轮巡时间	系统进行主备互换的轮巡时间。 缺省值：24 单位：小时
群控温度	群控时使用的温度类型。设置为“群组平均”

时间设置

在系统主界面上单击“设置”后选择“时间设置”页签，系统显示如图 4-15 所示的界面。在该界面可设置机组的当前系统时间。

设备出厂的默认时间设置为北京时间，请根据实际情况进行调整。

图 4-15“时间设置”界面



账号管理

在系统主界面上单击“设置”后选择“账号管理”页签，系统显示如图 4-16 所示。

图 4-16“账号管理”界面



操作员可通过账号设置功能修改客户密码。客户密码默认为“0001”，建议首次登录控制系统后进行修改。

4.8 维护

在系统主界面上单击“维护”，系统显示如图 4-17 所示的界面。

图 4-17 维护界面



MC90HDNC1L/R 一体式空调暂不支持当前维护界面操作功能，参数设置无效。

4.9 更多

在系统主界面上单击“更多”，根据提示输入客户密码后，可查看控制系统版本信息和进行恢复出厂操作。

图 4-18 更多界面



默认客户密码为“0001”。如果用户在控制系统界面上修改参数值前未输入过密码，则在单击对应参数的输入框时，系统自动弹出密码输入框。仅当密码输入正确后，才允许对参数值进行修改。

5 例行维护

5.1 概述

例行维护是一种预防性的维护，指维护人员在设备的正常运行过程中，周期性地开展检查和维护工作，及时发现并消除设备中的告警和故障隐患。

例行维护的目的是使设备能够长期稳定地运行，通过例行维护：

- 能及时的发现设备运行中的相关告警，避免告警进一步严重，保证设备的稳定运行。
- 能及时发现故障隐患，将故障消灭在萌芽期，避免故障发生后的经济损失和客户满意度的降低。
- 能发现设备运行趋势，有针对性地实施优化，提高设备的运行效率。

5.2 空调外观维护

空调外观的例行维护如表 5-1 所示。

表 5-1 空调外观维护

维护项	维护标准	维护周期	检测方式	处理方法
空调外观	空调清洁无尘、无污垢	6 个月	目测	使用棉布清除机组灰尘、污垢。

说明

以上维护周期仅为建议值，用户可根据实际规划进行调整。

5.3 电源线缆和电源端子维护

电源线缆和电源端子的维护如表 5-2 所示。

表 5-2 电源线缆和电源端子维护

维护项	维护标准	维护周期	检测方式	处理方法
接线面板电源线缆、电源端子的可靠性	电源线缆无松脱	3 个月	目测	断电后使用螺丝刀对松动线缆进行紧固。
	电源端子无松动	3 个月	目测	断电后使用螺丝刀对松动的电源端子重新固定。
	电源线缆无老化、破损、异常发热及其他异常	3 个月	目测	断电后更换电源线缆。
	接线面板处无灰尘	3 个月	目测	断电后使用毛刷清理干净灰尘。

说明

以上维护周期仅为建议值，用户可根据实际规划进行调整。

5.4 内外循环风机维护

内外循环风机的例行维护如表 5-3 所示。

表 5-3 内外循环风机维护

维护项	维护标准	维护周期	检测方式	处理方法
风机运行可靠性	风机无灰尘，风口处无异物堵塞	1 个月	目测	断电后使用毛刷或棉布清理干净风机的灰尘，并移除风口处的阻碍物。
	风机运行无异响	1 个月	目测	断电后使用螺丝刀对风机进行紧固。

说明

以上维护周期仅为建议值，用户可根据实际规划进行调整。

5.5 排水孔和换热器维护

注意

冷凝器和蒸发器的叶片边缘锋利，请穿戴手套进行维护操作。

排水孔和换热器的例行维护如表 5-4 所示。

表 5-4 排水孔和换热器维护

维护项	维护标准	维护周期	检测方式	处理方法
冷凝水排放	冷凝水排放口无堵塞。	6 个月	目测	断电后使用高压水枪冲洗冷凝水排放口。
冷凝器清洁	冷凝器无脏堵。	6 个月	目测	断电后冲洗冷凝器直至排出的水变清时，冲洗结束。
	翅片无严重变形	6 个月	目测	断电后用镊子或翅片梳等工具进行校正。
蒸发器清洁	蒸发器无脏堵。	12 个月	目测	断电后用干棉布擦除脏堵物。
	翅片无严重变形	12 个月	目测	断电后用镊子或翅片梳等工具进行校正。

说明

以上维护周期仅为建议值，用户可根据实际规划进行调整。

6 故障管理

6.1 故障维修

6.1.1 风机故障

风机类常见故障及维修方法如表 6-1 所示。

表 6-1 风机故障维修

故障现象	可能原因	检查和维修
内循环风机不运行	空调未上电	• 检查空调电源输入端是否有电。 • 检查空调外部断路器是否闭合。
	空调电源输入异常，如：电源过压、欠压、缺相	通过 RS485 通信或连接外接显示器，确定空调是否存在相应告警。
	空调属于待机状态	正常现象，空调上电，待机 30s 后，进入自动控制逻辑。
	风机卡死	检查是否有其他部件阻碍风机运行，或风机内存在异物。
	端子松脱	检查风机接线端子是否松脱。
	风机损坏	若存在蒸发器风机故障告警，且电源接线无异常，风机无异物卡住，则可能为风机损坏，需更换风机。
	主控板故障	若排除以上可能原因，请检测主控板是否故障。 主控板故障时请进行更换。
外循环风机不运行	压缩机未启动	压缩机启动后外风机才启动，且需要高压侧制冷剂到达一定的压力。
	风机卡死	检查是否有异物卡住风机。
	端子松脱	检查风机接线端子是否松脱。
	风机损坏	若冷凝器风机电源接线无异常，风机无异物卡住，则可能为风机损坏，需更换风机。
	主控板故障	若排除以上可能原因，请检测主控板是否故障。 主板故障时请进行更换。
风机异响	风机轴承磨损	更换风机。

故障现象	可能原因	检查和维修
	扇叶刮擦其它物体	检查是否有线缆等与风机扇叶干涉。
	风机固定螺栓松动	重新紧固风机的固定螺栓。

6.1.2 制冷系统故障

制冷系统类常见故障及维修方法如表 6-2 所示。

表 6-2 制冷故障维修

故障现象	可能原因	检查和维修
压缩机未启动	未开电源（待机）	检查主电源开关，并检查操作显示界面，是否已经开机。
	电路连接松动	紧固电路接头。
	电路开路或短路	检查电路开路或短路处，并维修总电源。
	变频器故障	更换变频器。
	控制板损坏	更换控制板。
	压缩机电机故障	更换压缩机。
压缩机不工作	控制温度设置过高，无制冷需求	检查进出风温度，并设置合适的控制温度。
排气压力高	冷凝器脏堵	清洁冷凝器。
	外循环风机不运转	参见表 6-1 确认冷凝器风机不运行现象处理措施。
	内循环风机不运转	参见表 6-1 确认蒸发器风机不运行现象处理措施
蒸发器冻结	内循环风机不运转	参见表 6-1 确认蒸发器风机不运行现象处理措施。

6.2 部件更换

6.2.1 更换内循环风机

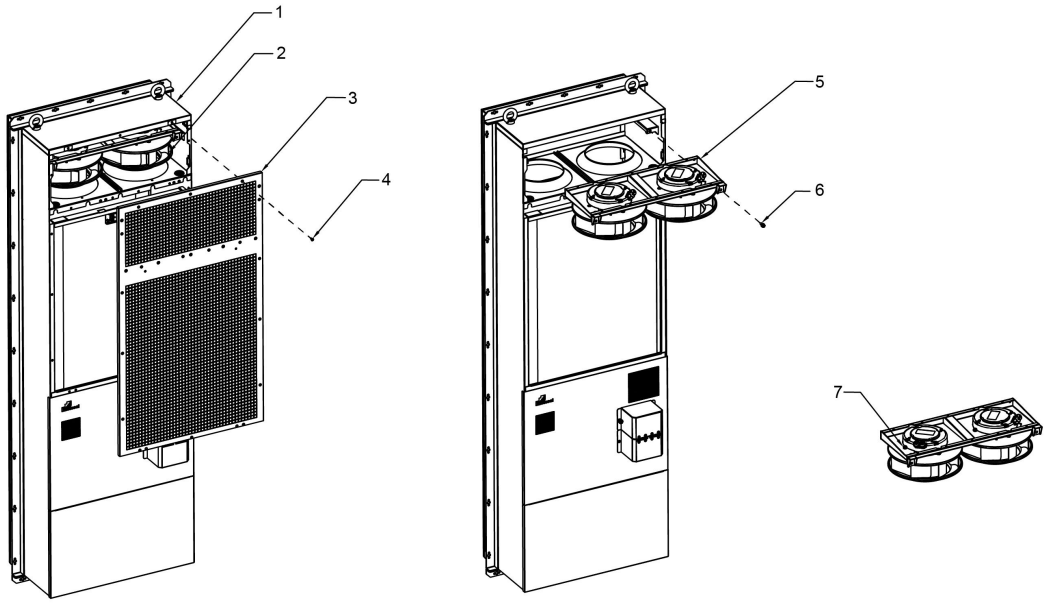
前提条件

- 已切断电源。
- 已准备好一字螺丝刀、十字螺丝刀和斜口钳。

背景信息

更换内循环风机的操作指引如图 6-1 所示。

图 6-1 更换内循环风机



（1）空调　（2）内循环风机　（3）前盖板

（4）前盖板螺钉（M4*10）　（5）内风机固定板　（6）内风机固定板螺钉（M6*16）

（7）内风机安装螺钉（M6*16）

操作步骤

步骤 1 使用十字螺丝刀，拆除内循环面板与机框螺钉(如图 6-1 中标识 3 所示,螺钉规格十字沉头 M4*10)。

步骤 2 使用斜口钳，拆除内风机线缆的固线扎带。

步骤 3 拆除内风机的接线端子。

步骤 4 使用十字螺丝刀，拆除内风机固定板螺钉(如图 6-1 中标识 6 所示，螺钉规格十字组合 M6*16)。

步骤 5 向外抽取出内风机与内风机固定板。

步骤 6 使用十字螺丝刀，拆除内风机与内风机固定板之间的螺钉(如图 6-1 中标识 7 所示，螺钉规格十字组合 M6*16)。

步骤 7 维护完成后按照相反顺序安装内循环风机。

6.2.2 更换外循环风机

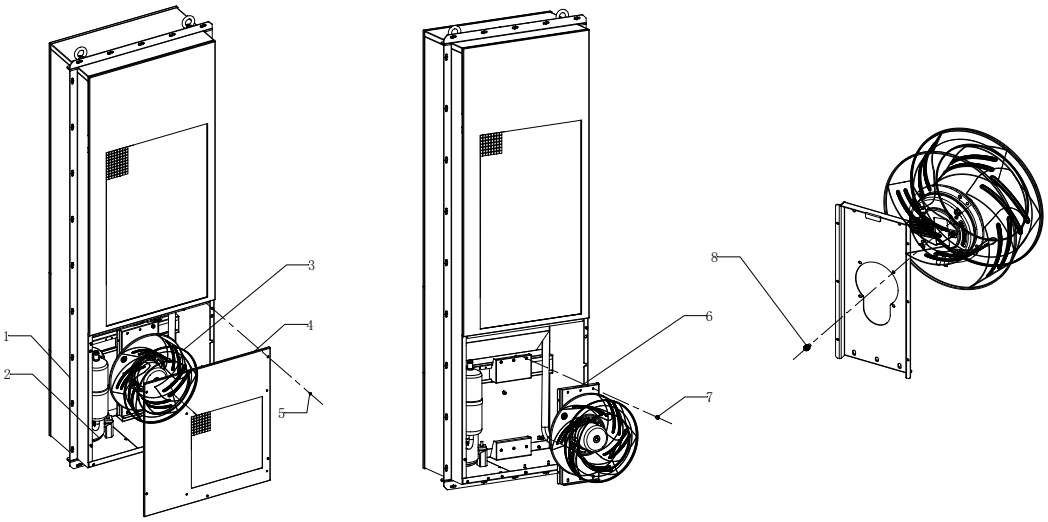
前提条件

- 已切断电源。
- 已准备好一字螺丝刀、十字螺丝刀、专用防盗螺丝刀和斜口钳。

背景信息

更换外循环风机的操作指引如图 6-2 所示。

图 6-2 更换外循环风机



（1）空调　（2）压缩机　（3）外风机

（4）外风机维护面板　（5）外风机维护面板螺钉（M4*16 防盗螺钉）　（6）外风机固定板

（7）外风机固定板安装螺钉（M6*16）　（8）外风机固定螺钉（M6*16）

操作步骤

步骤 1 使用专用防盗螺丝刀，拆除外风机维护面板(如图 6-2 中标识 4 所示，螺钉规格：M4*16 防盗螺钉)。

步骤 2 拆除外风机线缆的固定扎带。

步骤 3 拆除外风机的接线端子。

步骤 4 使用十字螺丝刀，拆除外风机的固定板螺钉(如图 6-2 中标识 7 所示，螺钉规格：M6*16)。

步骤 5 使用十字螺丝刀拆除外风机固定板，取出外风机进行更换(如图 6-2 中标识 8 所示，螺钉规格：M6*16)。

步骤 6 维护完成后按照相反顺序安装外风机。

6.2.3 更换压缩机

注意

压缩机故障时请联系厂家售后进行整机更换，严禁私自处理。

严禁将该故障的压缩机作为未分类的城市垃圾处理，必须遵循相关特殊要求独立处理。

6.2.4 更换控制板

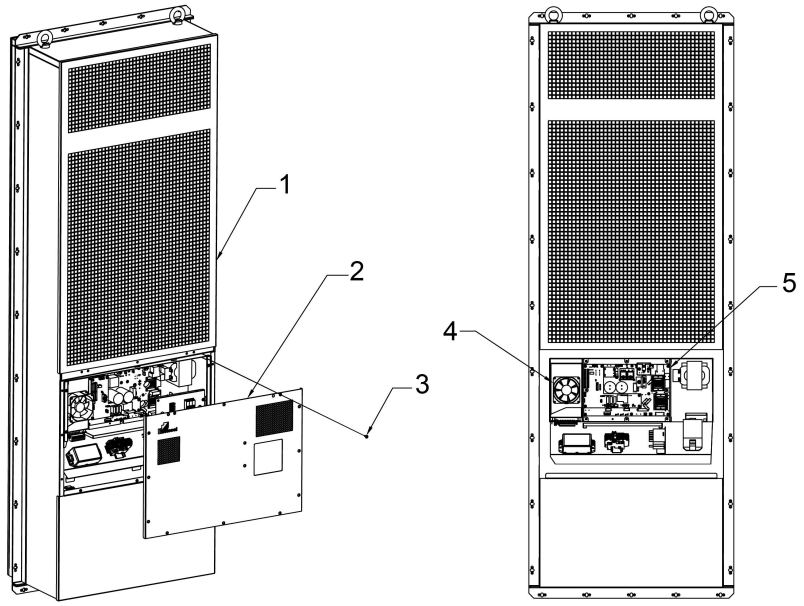
前提条件

- 已切断电源。
- 已准备好一字螺丝刀、十字螺丝刀和钳流表。

背景信息

更换控制板的操作指引如图 6-3 所示。

图 6-3 更换控制板



（1）空调　　（2）电控面板　　（3）电控面板螺钉（M4*10）

（4）散热风扇　　（5）驱控一体板

操作步骤

步骤 1 使用十字螺丝刀，拆除机组的维护面板(如图标识 3 所示，沉头螺钉规格：M4*10)。

步骤 2 拆除驱控一体板上所有的接线端子。

步骤 3 使用十字螺丝刀，拆除驱控一体板的固定螺钉(组合螺钉规格：M4*16)。

步骤 4 取出控制板进行更换。

步骤 5 维护完成后按照相反顺序安装。

电气端子的连接请参考空调的内部接线图。

说明

- 如果存在线缆破损等情况请一并进行更换。
- 其他电控器件的维修操作步骤如更换驱动板操作类似。

6.3 告警处理

锁定类告警（如下表中的“高压锁定告警”）需通过对机组先断电再上电的方式手工解除，其余告警在当前场景不满足告警条件时自动解除。

机组的告警信息如表 6-3 所示。

表 6-3 机组告警列表

告警类型	告警产生条件	系统缺省处理	排查方法	故障处理
回风高温告警	回风温度高于告警设定值	机组保持在制冷模式运行	1.检查高温告警设定值是否合理。 2.检查机组制冷量是否满足需求。	1.恢复系统的出厂设置，或拨打服务确认合理值后进行设置。 2.增加空调数量。
回风低温告警	回风温度低于告警设定值	机组保持在加热模式运行	1.检查低温告警设定值是否合理。 2.检查电加热是否故障。	1.恢复系统的出厂设置，或拨打服务确认合理值后进行设置。 2.电加热故障时进行更换。
回风高湿告警	回风湿度高于告警设定值	机组保持在除湿模式运行	1.检查回风湿度设定点或高湿告警设定点是否合理。 2.检查是否有门窗打开导致室内外空气流通。	1.恢复系统的出厂设置，或拨打服务确认合理值后进行设置。 2.关闭门窗并检查机组与安装面的密封措施是否正常。
回风温感故障告警	回风温度传感器与控制板通信中断或损坏	机组根据出风温感进行逻辑控制	1.检查回风温度传感器的电控接线是否，线缆是否损坏。 2.检查回风温度传感器是否故障。 3.检查控制板是否故障。	1.紧固连接端子或更换线缆。 2.更换传感器。 3.更换控制板。

告警类型	告警产生条件	系统缺省处理	排查方法	故障处理
内风机故障告警	内风机与控制板通信中断或损坏	机组停止运行	1.检查连接端子是否松脱，线缆是否损坏。 2.检查是否存在异物卡住风扇。 3.检查内风机是否故障。 4.检查控制板是否故障。	1.紧固连接端子或更换线缆。 2.清理异物。 3.更换内风机。 4.更换控制板。
系统高压告警	制冷剂的系统压力达到高压告警触发点	机组停止运行	1.检查环境温度是否过高。 2.检查高压开关接线是否松动，或线缆损坏导致误告警。 3.检查冷凝器是否严重脏堵。 4.检查回风口是否存在杂物堵塞。 5.检查冷凝器风机是否断电或工作异常。	1.根据安装要求正确摆放机组，避免让机组受阳光直射。 2.紧固高压开关的接线或更换线缆。 3.清洗冷凝器。 4.清理机组回风口前的脏堵物。 5.非断电原因时更换冷凝器风机。
高压锁定告警	1 小时内 3 次触发系统高压告警	机组停止运行	1.检查环境温度是否过高。 2.检查高压开关接线是否松动，或线缆损坏导致误告警。 3.检查冷凝器是否严重脏堵。 4.检查回风口是否存在杂物堵塞。 5.检查冷凝器风机是否断电或工作异常。	1.根据安装要求正确摆放机组，避免让机组受阳光直射。 2.紧固高压开关的接线或更换线缆。 3.清洗冷凝器。 4.清理机组回风口前的脏堵物。 5.非断电原因时更换冷凝器风机。
系统低压告警	制冷剂的系统压力达到低压告警触发点	机组停止运行	1.检查制冷剂是否存在泄漏。 2.检查电子膨胀阀是否故障。	1.检查泄漏点，维修后补充制冷剂。 2.更换电子膨胀阀。

告警类型	告警产生条件	系统缺省处理	排查方法	故障处理
低压锁定告警	1 小时内 3 次触发系统低压告警	机组停止运行	1.检查制冷剂是否存在泄漏。 2.检查电子膨胀阀是否故障。	1.检查泄漏点，维修后补充制冷剂。 2.更换电子膨胀阀。
交流过压告警	输入交流电压高于运行电压范围	机组停止运行	1.检查机组供电电压。 2.检查驱动板上的防雷器件是否故障。	1.检查供电电网是否稳定。 2.防雷器件故障时进行更换。
交流欠压告警	输入交流电压低于运行电压范围	机组停止运行	1.检查机组供电电压。 2.检查驱动板上的防雷器件是否故障。	1.检查供电电网是否稳定。 2.防雷器件故障时进行更换。
异常掉电	开机状态下交流输入断开	机组断电关闭	检查供电电网是否断电。	检查供电电网。
吸气温感故障	传感器损坏或通信中断	机组停止运行	1.检查吸气温度传感器的电控接线是否松动，线缆是否损坏。 2.检查吸气温度传感器是否故障。 3.检查控制板是否故障。	1.紧固电控接线或更换线缆。 2.更换吸气温度传感器。 3.更换控制板。
盘管防冻保护	蒸发器盘管温度低于-2℃	机组停止运行	1.检查过滤网是否脏堵，导致风量偏小。 2.检查蒸发器是否脏堵影响散热。	1.清洁过滤网。 2.清洁蒸发器。

7 参考

空调的条形码样例如图 7-1 所示。

图 7-1 空调条形码样例



- ①：厂家自定义（M01XXXX***XXXX）。
- ②：生产日期，如 21621 表示 2021 年 6 月 21 日。A/B/C 表示 10 月/11 月/12 月。
- ③：本批次产品的序列号。

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厂家对产品持续升级进行的资料更改，恕不另行通知。

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